

AI-Powered Smart Waste Sorting Bin (SORT E-BIN)

Capstone Project Report

END SEMESTER EVALUATION

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ABSTRACT

The rapid pace of urbanization and industrial growth has made solid waste management a critical environmental challenge. Research shows that improper segregation at the source remains one of the biggest barriers to effective recycling, often leading to contamination of recyclable streams, inefficient recovery, and increased landfill dependency. While prior works have demonstrated the potential of AI-driven waste classification, IoT-based monitoring, and smart garbage bins, most remain limited to simulations or partial implementations, lacking full integration into a deployable system.

To overcome these limitations, the **SORT E-BIN** project introduces a complete, intelligent waste segregation solution that combines Convolutional Neural Networks (CNN), IoT-enabled monitoring, and automated sorting mechanisms. The system employs an InceptionResNetV2 model for high-accuracy classification of waste into categories such as plastic, paper, metal, glass, and organic matter. Once identified, servo motors direct the waste to the appropriate compartment. Designed for deployment in high-footfall campus areas, SORT E-BIN demonstrates strong potential for scalability to urban smart city infrastructures. By integrating AI, IoT, and robotics into a single modular and energy-efficient unit, this project contributes a practical and sustainable approach to one of the most pressing challenges in modern waste management.

DECLARATION

We hereby declare that the design principles and working prototype model of the project entitled **AI-Powered Smart Waste Sorting Bin (SORT E-BIN)** is an authentic record of our own work carried out in the Computer Science and Engineering Department, TIET, Patiala, under the guidance of Dr. Ravneet Kaur and Dr. Harcharan Jit Singh during 7th semester (2025).

Date: 23-12-2025

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LIST OF ABBREVIATIONS

Abbreviation Full Form

AI-	Artificial Intelligence
AIoT-	Artificial Intelligence of Things
CNN-	Convolutional Neural Network
IoT-	Internet of Things
MSW-	Municipal Solid Waste
RGB-	Red Green Blue (image color model)
GPU-	Graphics Processing Unit
DNN-	Deep Neural Network
ReLU-	Rectified Linear Unit
IEEE-	Institute of Electrical and Electronics Engineers
CVPR-	Computer Vision and Pattern Recognition
ICIICC-	International Conference on Innovative Computing and Communications

CHAPTER 1 | INTRODUCTION

1.1 Project Overview

Waste management has emerged as one of the most significant environmental and public health challenges due to rapid urbanization, industrial growth, and increasing consumption. The sharp rise in municipal solid waste (MSW) generation, combined with inefficient disposal and segregation practices, has led to environmental pollution, health risks, and increased landfill dependency [1]. Improper handling of waste contaminates recyclable materials, reduces recycling efficiency, and increases operational costs for waste management authorities [2].

A major limitation of existing waste management systems is their dependence on manual segregation and user compliance. Despite the availability of categorized bins, incorrect disposal at the source is common due to lack of awareness and inconsistent human behavior [3]. Even minor contamination of recyclable waste can render entire batches unsuitable for recycling, resulting in material loss and processing inefficiencies [4].

The SORT E-BIN project addresses this challenge by introducing an AI-powered smart waste sorting system that integrates Artificial Intelligence (AI), Internet of Things (IoT), and automated mechanical sorting. This system employs a deep learning-based image classification model deployed on an edge device to identify waste categories in real time [5]. Based on these classification output, a servo-motor-driven mechanism physically directs the waste into the appropriate compartment, ensuring accurate segregation at the point of disposal.

In addition to automated classification and sorting, SORT E-BIN incorporates IoT-enabled monitoring using ultrasonic sensors to track bin fill levels. The collected data enables real-time monitoring and optimized

waste collection planning, reducing overflow and operational inefficiencies [6]. Designed with scalability, sustainability, and modularity in mind, this system is suitable for deployment in institutional environments such as college academic departments and can be extended to larger urban offices.

1.1.1 Technical Terminology

First, in understanding the technical foundation for the SORT E-BIN system, following are some key terms and technologies that are core to the project:

- Refers to all waste originating from residential, commercial, and institutional sources. Components of MSW include plastics, paper, metals, and glass.
 - AI is the ability of machines to make representations of processes similar to those of human intelligence, including learning, reasoning, and decision-making. In this paper, AI is applied to classify waste automatically.
 - **Convolutional Neural Network:** This architecture of deep learning that is particularly effective for image recognition and classification tasks. CNNs automatically extract the hierarchical features from images, which are the best fit to find the difference between various categories of waste.
 - **Transfer Learning:** The concept of transfer learning in machine learning is a model pre-trained on a large dataset, say ImageNet, that gets fine-tuned to perform a certain task. Transfer learning in the case of SORT E-BIN improves the classification accuracy with limited domain-specific data.
- Edge Computing:** The processing of data without the use of cloud servers on active embedded hardware itself; the AI model in SORT E-BIN executes on edge devices like Raspberry Pi, thus enabling low-latency and real-time model inference.
- **Internet of Things IoT:** A network connecting devices that have sensors as well as other forms of communication. IoT enables real-time bin fill levels and system status monitoring.
 - **Servo Motors:** Used to operate the movement mechanisms with precision, usually used in waste management technologies. Servo motors Sort E-Bin guide the waste into the appropriate compartment.

1.1.2 Problem Statement

Waste management has become one of the most critical environmental and public health issues of the 21st century. The surge in rapid urbanization, extension of industry, and improvement in the standard of living have resulted in increased consumption and thereby increased generation of municipal solid waste at an unprecedented rate. Global studies have estimated that about 2.01 billion tonnes of MSW were generated in 2016, which is projected to increase to 3.40 billion tonnes by 2050 if business-as-usual scenarios continue. Such high growth puts immense pressure on existing waste management infrastructure.

Contributing to this problem is a lack of proper segregation at the source itself. In fact, in many urban areas and institutions, color-coded or category-based separation for materials like plastic, paper, metal, and glass are in use. Proper segregation hence relies largely on the awareness and participation level of users themselves. Many times, these are not practiced due to time, knowledge, or interest factors, thus contaminating the 'recyclable' and 'non-recyclable' materials together.

Among all the critical challenges of modern recycling systems, contamination of recyclable waste plays an important role. A single non-recyclable or organic item mixed into the stream of recyclable waste can render a whole batch unsuitable for processing. This leads to increased landfill dependency, wastage of recoverable materials, and additional operational costs for the facility. In some cases, contamination leads to the mechanical damage of recycling equipment, thus increasing the costs even further.

Large volumes of diverse waste generated daily in high-footfall environments such as college campuses, shopping complexes, and public institutions add to the problem. Manual segregation in such places is time-consuming, inefficient, and dangerous to the health of the sanitation worker. Most conventional waste bins also lack real-time monitoring; this results in inefficient collection schedules, overflowing bins, unhygienic conditions, and a waste of fuel for unnecessary rounds during collections.

Therefore, the basic problem to solve is the lack of an intelligent, automated, and scalable solution to segregate waste at the point of disposal correctly while enabling real-time monitoring and data-driven waste management.

1.1.3 Goal

The prime objective of the SORT E-BIN project is to design and develop an **AI-based smart sorting system** for waste, which will help to classify and separate the waste at the source level, thus resulting in minimal contamination and efficient recycling.

This project aims to achieve, specifically:

- To properly classify various forms of waste like plastic, paper, and metal through image classification techniques based on deep learning.
- To separate waste physically into respective compartments through an automated mechanical sorting system.
- For real-time monitoring of bin fill levels by means of IoT-based sensors.
- To minimize human interaction during waste segregation, hence reducing operation costs and health associated with the process.
- To create a scalable and energy-efficient approach that is implementable in the institutional setting.

With the accomplishment of these objectives, SORT E-BIN hopes to fill the gap that exists between research-grade models of Artificial Intelligence and effective, reality-based waste management systems.

1.1.4 Solution

In an attempt to rectify the issue of improper waste segregation and inefficient waste management techniques, the SORT E-BIN project suggests the implementation of an AI-driven automated waste segregation system entailing the synchronization of deep learning, edge computing, IoT-related observation, and mechanical actuators.

At the core of the system, there lies the development of the waste classification system using a Convolutional Neural Network (CNN) and the power of deep learning, where transfer learning concepts are employed to train the InceptionResNetV2 architecture using a combination of publicly available datasets like TrashNet and custom collected images related to waste. It enables the system to learn distinguishing features related to different types of wastes like plastic, paper, and metal. Transfer learning increases the accuracy and efficiency of the process.

The trained CNN is run on an edge computing platform like a RaspberryPi. Edge computing enables real-time processing with zero latency and cloud independence,

making this suitable for indoor and semi-controlled settings like an educational campus. Upon addition of a waste material into the bin, the camera module captures the picture and transmits it to the edge device, which classifies it via AI in less than a second.

On the basis of the prognosticated type of waste, a mechanical sorting unit using servo motors is triggered. The servo motors turn and tilt a disc structure to mechanically guide the waste category into its respective container. By means of such mechanical activation, automatic segregation at the point of origin takes place with lesser chances of contamination between waste categories and without requiring manual segregation at a later stage. The mechanical design is readily maintainable and expandable for segregating further waste categories in subsequent models.

Apart from the automated classification sorting system, the system also has an IoT-based monitoring system. Ultrasonic sensors are employed in every compartment of the waste bins to measure their level continuously. The data obtained from ultrasonic sensors is sent to the monitoring system by the edge device. This allows real-time usage tracking of the bins by the authorities. This has various advantages, from optimized collection times to the prevention of overflows.

The design focuses on efficiency, reliability, and scalability. SORT E-BIN uses a combination of AI vision technology, automation by machinery, and IoT sensors in order to provide an implementable solution to smart trash separation. It can be ideally used in high traffic institutional areas like college campuses where a huge amount of mixed trash is collected on a daily basis. In conclusion, the SORT E-BIN system enables the transformation of traditional waste bins into smart systems for the automation and classification of waste and its monitoring in real-time. The system thus provides a crucial solution to the problems associated with the segregation of waste and can be considered important for effective waste recycling.

1.2 Need Analysis

Waste management has now emerged as a pressing global issue, but in dense volume areas like campus setups, the need for efficient waste management systems has escalated. Various types of wastes are generated in a college campus, including paper,

plastic, and organic waste, along with other recyclables like electronics and hazardous wastes, making segregation in such scenarios both difficult and unavoidable [2]. The process of manual segregation of waste in rubbish trucks has been found not only laborious and time-consuming but also highly vulnerable to human errors. This can lead to contamination of recyclables with other types of wastes, thereby reducing their recyclability and dependence on landfills [3].

Contamination is one of the major factors hindering efficient recycling. In particular, the mixture of plastics, papers, and metals with waste foods or non-recyclable items makes the whole waste unfit for recycling [2][4]. In a college setting, the mixture of leftover foods, packaging papers, and stationery becomes common in the waste bins as a result of a poorly defined waste disposal system and inefficient labeling. It has been observed in research that poorly defined segregation rules and a lack of engagement of the end-users result in the indiscriminate dumping of waste, making the whole waste recycling inefficient and a waste of opportunities [5]. Furthermore, the issue is exacerbated by a lack of knowledge of waste disposal rules among the campus community members [6]. Moreover, the high frequency of waste generation in busy zones on campus generates massive quantities of mixed waste daily, leading to bottlenecks in waste management and recycling on campus [1][3].

Added to this is the fact that currently, there is no real-time monitoring of waste data to inform the decision-making processes in most waste management systems in institutions. As such, when the levels of waste within the bins are not monitored properly, waste collection schedules often become inefficient, resulting in full bins in correspondence to traffic rates while others remain half-full [4][7]. This results not only in unhygienic practices, considering that waste is not disposed of as needed, but also results in an increase in greenhouse gas emissions as a consequence of wastage collection [8].

A smarter, automated waste segregation system, therefore, is the need of the hour to overcome this problem. SORT E-BIN specifically aims to satisfy this requirement by combining the capabilities of AI-powered image recognition, mechanical sorting using a servo motor, and IoT technology tracking, all under one compact device [2, 5]. SORT E-BIN, therefore, prevents the phenomenon of cross-contamination and decreases labor

by performing accurate segregation of wastes right from the disposal phase [6]. IoT-based SORT-E BIN also helps monitor the status of bin filling levels and statistics of wastes disposed of, and optimization regarding routing and scheduling can thus be made using this device [7, 8]. An interactive display provided by SORT E-BIN can also instruct people regarding disposal, leading to a change in behavior regarding proper disposal practices among university students and faculties [3, 9]. If implemented in places with heavy footfall in the campus, like the hostels, Jaggi, Kravings, COS, and G Block, SORT E-BIN will enable an active and innovative approach to sustainability for the respective institutions. Not only will this help in reducing waste sent to landfills and increase recycling efficacy, but also create environmentally responsible awareness in the campus ecosystem. With the rising initiative from the government and international bodies for sustainable waste management practices, AI-assisted sorting systems, like SORT E-BIN, have truly become the need of the hour for those institutions aiming towards long-term environmental sustainability [4, 9].

1.3 Research Gaps

Despite considerable advancements in waste management technologies, there remain significant gaps in research and implementation that limit the efficiency and scalability of intelligent waste segregation systems. Current literature in the domain of smart garbage classification and AIoT-based waste management reveals that while progress has been made in computer vision models, IoT integration, and prototype smart bins, the majority of solutions are either limited to lab-scale testing, lack real-world deployment readiness, or address only part of the overall problem. The SORT E-BIN project directly targets these shortcomings, aiming to provide a fully integrated, deployable, and user-friendly waste sorting solution.

1. Limited Handling of Real-World Multi-Object Scenarios

Most existing systems assume a clean, single object is presented to the camera for classification. For example, images are taken in controlled environments where items are isolated against a plain background [4]. However, in real college or public settings, waste is often thrown in groups, with items overlapping or partially hidden. Current models are not designed to handle such clutter, which reduces classification accuracy. A research gap

exists in developing AI models capable of robust multi-object detection and recognition in messy, “in-the-wild” bin conditions [5].

2. **Small and Non-Representative Training Datasets**

Many of the models in the reviewed papers are trained on datasets like TrashNet or limited custom collections (e.g., only a few thousand images). These datasets do not capture the full variability of real waste — such as crumpled wrappers, food-soiled paper, transparent plastics, or items under poor lighting. This limits generalisation to real-world scenarios. There is a clear need for large, diverse, campus-specific datasets that better represent the complexity of daily waste streams [3][6].

3. **Over-Reliance on Controlled Image Capture Pipelines**

Several research prototypes acquire images under fixed lighting and controlled angles (e.g., top-down views with uniform backgrounds) [2][4]. While this improves classification performance in labs, it does not reflect real usage conditions where bins are used by hundreds of people in varying environments. Robust systems must be tested against variable illumination, camera vibration, blurred images, and random object orientations, conditions largely ignored in current studies [5][7].

4. **Limited Evaluation of Mechanical Sorting Systems**

Though many papers describe how servos or rotating bases can move waste into compartments, very few evaluate long-term reliability, jam prevention, or safety considerations. For instance, how does the system handle oversized or irregularly shaped items? What if waste blocks the sorting mechanism? Research has not sufficiently addressed maintenance, durability, and safe interaction in high-footfall areas. This gap is crucial for ensuring real-world usability.

5. **Insufficient Exploration of Edge AI Deployment Constraints**

While some works have tested lightweight models like MobileNet or EfficientDet on Raspberry Pi [6][9], there is little analysis of long-term performance under real loads. Key concerns such as power consumption, thermal throttling, latency when multiple items are processed, or running multiple sensors simultaneously are underexplored. A research gap exists in systematically evaluating the trade-offs between accuracy, speed, and energy efficiency in edge-based deployments.

6. **Over-Dependence on Cloud Processing**

Several systems rely on cloud platforms to process classification results and manage data. While effective for monitoring [7][10], this introduces risks of latency, internet dependency, and data privacy concerns. For deployment in institutions or public spaces, a balance is needed between edge computing and cloud integration. Current research has not sufficiently explored hybrid models that are both privacy-preserving and reliable.

1.4 Problem Definition and Scope

Problem Definition

The growing challenge of municipal solid waste (MSW) management has reached critical levels, with global waste generation expected to rise from 2.01 billion tonnes in 2016 to 3.40 billion tonnes by 2050 [1]. A major contributor to this crisis is the lack of effective segregation at the source, which results in recyclables being contaminated by food and non-recyclables. Once contaminated, otherwise valuable materials such as plastics, paper, and metals often end up in landfills, increasing environmental degradation and economic losses [2].

Research highlights that although efforts have been made to introduce separate bins for different waste categories, user non-compliance, limited awareness, and inconsistent disposal habits continue to hinder success. In educational institutions and other high-footfall areas, diverse waste streams such as food leftovers, packaging, stationery, and plastics are frequently mixed together. This not only reduces recycling efficiency but also burdens collection teams with additional manual sorting, exposing them to health risks and operational inefficiencies [5][6].

Moreover, the absence of real-time monitoring and predictive collection strategies leads to overflowing bins, inefficient collection routes, and unhygienic conditions on campuses and in urban spaces. Current research has made progress in developing AI-based waste classification models, but most systems remain confined to controlled laboratory conditions and do not address the complexities of real-world waste management, such as varied object shapes, contamination, and mechanical reliability[4].

Thus, the central problem lies in the gap between research prototypes and practical, scalable waste segregation solutions. Without automation that is accurate, efficient, and adaptable to real conditions, institutions and cities will continue to face increasing landfill dependency, higher carbon emissions, and serious public health challenges.

Scope of the Project

The scope of SORT E-BIN is to design and implement an **AI-powered smart waste segregation system** that integrates deep learning, IoT, and mechanical automation for real-time classification and disposal of waste. The system is being developed with the following components:

1. Software and AI Model Development

- The core of the system is a Convolutional Neural Network (CNN) built using InceptionResNetV2, a deep learning architecture pre-trained on ImageNet and fine-tuned for multi-class waste classification [6].
- The dataset is organised into train, validation, and test splits, and preprocessing includes corruption removal, resizing to 256×256 pixels, and application of augmentation techniques such as shear, zoom, and horizontal flip to improve model robustness in real-world conditions [3].
- A two-stage training strategy is employed:
 - Stage 1: Freeze the backbone network and train the classification head to learn waste-specific features.
 - Stage 2: Unfreeze the last 180 layers of the backbone for fine-tuning with a low learning rate, improving feature extraction for waste categories [6].
- To improve generalization, the training process incorporates callbacks including EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint, ensuring stability and optimal model performance.
- The final trained model is exported in.keras format for deployment on Raspberry Pi, enabling real-time inference on edge devices.
- Performance evaluation includes accuracy and loss curves, confusion matrix, and classification reports, ensuring transparency and reliability of the model.

2. Mechanical Sorting System

- The system uses a servo-motor-driven mechanism to physically direct waste into dedicated compartments based on AI classification [2].
- A modular compartment design allows separation of three major categories (plastics, paper, metal).
- The framework is designed for durability, ease of maintenance, and scalability, allowing future expansion for additional waste categories.

3. IoT-Enabled Monitoring and Data Analytics

- Each compartment is fitted with ultrasonic sensors to measure fill levels [3].
- Data is transmitted via Raspberry Pi to a cloud-based or local IoT dashboard [5].
- The dashboard enables administrators to monitor bin status in real time, track waste type distribution, and generate analytics for waste generation patterns.
- Predictive analytics from collected data will support optimized collection routes, reduced operational costs, and prevention of overflow conditions [4].

4. Hardware Components

The physical system integrates the following components:

- Raspberry Pi– Edge device for AI inference and IoT connectivity.
- Camera Module – Captures waste images for classification.
- Servo Motors – Operate the sorting mechanism.
- LED Display – Provides user feedback on classification results.
- Physical Compartments – Separate bins for different categories of waste.

5. Deployment and Evaluation

- Pilot deployment is planned in Thapar Institute at high-traffic zones such as hostels, Jaggi, canteens, COS, Kravings, and G Block, ensuring exposure to varied and high-volume waste streams.
- During the pilot phase, data will be collected on classification accuracy, fill-level monitoring, and user interaction.
- Feedback will be used to further refine the AI model, mechanical design, and IoT dashboard for greater reliability.

1.5 Assumptions and Constraints

Assumptions

1. Controlled Deployment Environment

- The system is assumed to be deployed in semi-controlled environments such as college hostels, canteens, and academic blocks, where lighting and space conditions are relatively stable [3].
- Images captured by the camera will have sufficient clarity for the CNN model to perform accurate classification [6].

2. Dataset Representativeness

- The training dataset is assumed to cover the major categories of waste generated on campus (plastic, paper, metal) [4].
- It is assumed that newly encountered waste items in deployment will closely resemble categories represented in the training dataset [6].

3. Reliable Power and Connectivity

- The system assumes access to a stable power supply for Raspberry Pi, servo motors, and sensors [5].
- Wi-Fi connectivity is assumed to be available for IoT data transmission and dashboard access [2].

4. User Interaction Compliance

- It is assumed that users will place only one item at a time into the bin, allowing the camera to capture clear images for accurate classification [6].
- Users are assumed to follow basic guidelines, e.g., not disposing of liquids directly or physically obstructing the camera/sensor [3].

5. Limited Waste Categories

- The initial prototype is assumed to segregate waste into 4 major categories only, with future scope for expansion to additional categories [4].

Constraints

1. Hardware Limitations

- The system is constrained by the processing capability of Raspberry Pi 4, which may limit the use of very large or complex deep learning models [5].
- Mechanical sorting is limited by the capacity and torque of servo motors, restricting the system to lightweight and moderately sized waste items [2].

2. Model Generalization

- The InceptionResNetV2-based model may struggle with unusual or ambiguous waste items (e.g., multi-material packaging, soiled items) due to limited dataset diversity [6].
- Performance may degrade under poor lighting or when waste items are partially occluded [3].

3. Scalability Challenges

- The prototype is designed for campus-scale deployment and is not yet optimized for large-scale municipal waste management [1].
- Expanding to outdoor public spaces would require weatherproofing, rugged design, and independent power sources [4].

4. Maintenance Requirements

- Sensors, servo motors, and the mechanical framework require periodic maintenance to ensure reliability [2].
- Misalignment of compartments or mechanical jams may disrupt accurate sorting [4].

5. Data Privacy and Storage

- The system captures waste images for classification; hence, data storage and transmission must comply with privacy and ethical guidelines, especially in public settings [7].
- Cloud storage limitations or downtime may affect IoT dashboard functionality.

1.6 Standards

- **IEEE 802.15.4** – This standard defines low-rate wireless personal area networks (LR-WPANs). It is widely used in IoT-based communication for devices such as sensors and microcontrollers. SORTIBIN will rely on this standard for reliable wireless communication between the smart bin, sensors, and cloud systems.
- **ISO 14001:2015** – This is an international standard for environmental management systems (EMS). It ensures that the design and deployment of SORTIBIN comply with sustainable waste management practices and help reduce environmental impact.
- **IEC 61508** – Functional safety of electrical/electronic/programmable electronic systems. This standard is essential for ensuring the safety of automated systems, particularly the robotic sorting mechanism in SORTIBIN.
- **IEEE 802.11 (Wi-Fi)** – This standard defines the protocol for wireless networking. It will be used to ensure seamless connectivity of SORTIBIN with cloud servers for real-time monitoring and updates.
- **ISO/IEC 27001** – This international standard provides guidelines for information security management systems. It is crucial for ensuring the security and privacy of the data collected by SORTIBIN (waste type logs, user activity, and IoT sensor data).

1.7 Approved Objectives

1. Automated Waste Classification

The system uses a Convolutional Neural Network (CNN) based on InceptionResNetV2, a pre-trained deep learning model, to automatically identify and classify waste into categories such as plastic, paper, metal[3][6]. The model is trained on a large dataset and optimized for real-time performance on the Raspberry Pi [4], ensuring fast and accurate classification of waste items at the source.

2. Intelligent Waste Sorting

Once the waste is classified, the result is used to control a servo-motor-based sorting mechanism. This directs the item into the correct compartment of the bin without requiring manual effort [2]. The design is modular, allowing for the

segregation of at least 4–5 waste categories, with the option to expand in the future [5].

3. Real-Time Monitoring & IoT Integration

The bin is equipped with ultrasonic sensors in each compartment to monitor fill levels. This data is transmitted via the Raspberry Pi to a cloud-based IoT dashboard [7], enabling administrators to track usage in real time, optimize collection schedules, and avoid overflow situations [1].

4. Energy-Efficient & Sustainable Design

The AI model is optimized for low-power edge computing, reducing computational load [4]. The system can also be integrated with solar panels for outdoor deployment, ensuring energy efficiency and sustainability [5].

5. User Engagement & Smart City Integration

The bin includes a user feedback display, which shows the detected waste category, raising awareness among users [6]. Future versions may include gamification features such as reward points for correct disposal. The system is also designed for compatibility with smart city waste management platforms, supporting large-scale sustainability initiatives [7].

1.8 Methodology

1. Data Collection and Preprocessing

- A diverse dataset of waste images (plastic, paper, glass, metal) was collected and organized into training, validation, and testing sets [3].
- Data cleaning was performed by removing corrupted or unreadable images [6].
- Data augmentation techniques (rotation, shear, zoom, flips) were applied to increase dataset diversity and improve model generalization [3][4].
- All images were resized to 256×256 pixels and normalized for input into the deep learning model [6].

2. Model Development

- A dataset of waste images was collected and preprocessed. Various models (InceptionResNetV2, AlexNet, MobileNet, VGG19) were trained, and their accuracy was compared.
- A Convolutional Neural Network (CNN) based on InceptionResNetV2 was selected for classification [6].

- The model was first used as a feature extractor with pre-trained ImageNet weights [4].
- Fine-tuning was later applied to the last 50 layers to improve performance on waste-specific data [6].
- The final model consists of layers for Global Average Pooling, Dropout (0.5), Dense (128 neurons with ReLU activation), and Softmax output for multi-class classification [3].
- Training was performed with Adam optimizer, using callbacks such as Early Stopping, ReduceLROnPlateau, and ModelCheckpoint to avoid overfitting and preserve the best weights [6].

3. Model Evaluation and Testing

- The trained model was evaluated on the test dataset, achieving high accuracy in correctly classifying waste items [4].
- Confusion matrices and classification reports were generated to analyze misclassifications and ensure balanced performance across all categories [6].
- A real-time prediction function was developed to classify single waste items at the point of disposal [3].

4. Hardware Integration

- The Raspberry Pi serves as the central processing unit, hosting the trained model for on-device inference [4].
- Camera module captures image of waste items as they are disposed into the bin.
- Based on the classification result, servo motors control mechanical flaps that direct waste into the correct compartment.
- Ultrasonic sensors are embedded in each compartment to measure fill levels.

5. IoT and Real-Time Monitoring

- Sensor data (bin fill levels) is collected and transmitted to a cloud-based dashboard via Wi-Fi [7].
- Administrators can view real-time bin status, receive alerts for full bins, and optimize waste collection schedules [1].

1.9 Project Outcomes and Deliverables

The implementation of SORT E-BIN is expected to produce measurable outcomes and tangible deliverables that demonstrate the system's technical functionality, efficiency, and impact on sustainable waste management.

1. **Automated Waste Classification** – A trained and deployed **InceptionResNetV2 CNN model** capable of classifying waste into categories such as plastic, paper and metal with high accuracy in real-time [3][6].
2. **Automated Sorting Mechanism** – A fully functional **servo-motor-based sorting system** integrated with the AI model, ensuring correct disposal of waste into designated compartments without manual intervention [2][5].
3. **Sustainability and Efficiency** – An energy-efficient system optimized for **edge computing** on Raspberry Pi with potential solar integration, contributing to sustainable smart city practices [4][5].
4. **Scalable Smart Waste Management Model** – A deployable prototype that can be replicated across multiple locations such as hostels, canteens, and academic buildings, creating a model for broader urban adoption [6][7].

1.10 Novelty of Work

The proposed solution is unique as none of the existing systems provide a complete end-to-end platform for automated waste segregation the way SORT E-BIN does. While research has explored AI-based waste classification and IoT-based monitoring separately, very few solutions combine AI-driven image recognition, hardware-based sorting, real-time IoT monitoring, and user engagement into a single deployable unit. Our system stands out by using an advanced InceptionResNetV2 CNN model deployed on Raspberry Pi, making it efficient, accurate, and easy to integrate without requiring heavy cloud infrastructure. The addition of servo-motor-based mechanical sorting ensures that classification directly results in physical waste segregation, something most competitors lack.

This combination of AI, IoT, hardware integration, and user awareness makes SORT E-BIN a truly novel and practical solution to one of today's most pressing challenges—waste management.

CHAPTER 2 | REQUIREMENT ANALYSIS

2.1 Literature Survey

2.1.1 Related Work

Waste management has become a critical global concern due to rapid urbanization and industrialization, which have significantly increased the generation of mixed municipal solid waste. Conventional manual waste segregation methods are inefficient, labor-intensive, and highly prone to human error. To address these limitations, researchers have explored the application of Artificial Intelligence (AI), particularly Convolutional Neural Networks (CNNs), for automated waste classification, as CNNs have demonstrated strong performance in image recognition tasks [1][2].

Fatimah Balqis et al. [1] developed *TrashBot*, an IoT-based waste segregation system that employed SSD MobileNet V2 and EfficientDet-D0 models for object detection. The system was trained on a dataset of 2401 images covering plastic, paper, and metal categories and deployed on Raspberry Pi platforms. While SSD MobileNet V2 achieved reasonable precision and recall, EfficientDet-D0 was found to be computationally expensive for real-time edge deployment, highlighting the trade-off between accuracy and processing efficiency on resource-constrained devices.

Aviral Jain et al. [2] proposed a smart garbage classification system integrating EfficientNet-b0 with IoT infrastructure. Their prototype utilised a conveyor-based segregation mechanism and demonstrated high classification accuracy across multiple waste categories. The study emphasised the effectiveness of transfer learning in improving model performance for real-world garbage datasets.

Puspaningrum et al. [3] presented an AIoT-based waste classification system using lightweight CNN architectures optimised for low-power embedded devices. Their work focused on reducing computational cost without significantly compromising accuracy, demonstrating that AI models can be effectively deployed on resource-constrained IoT hardware for scalable waste segregation.

Hossain et al. [4] explored the role of Explainable Artificial Intelligence (XAI) in smart city applications, including waste management. Their work highlighted the importance of transparency and interpretability in AI-driven public systems to ensure trust and

reliability, particularly in environmental monitoring and automated decision-making systems.

Lee et al. [5] investigated object sorting for smart garbage applications by combining computer vision techniques with robotic actuation. Their study analysed challenges such as overlapping objects and partial occlusion in mixed waste scenarios and recommended improvements in dataset diversity and mechanical robustness to enhance real-world performance.

Recent studies have also explored YOLO-based approaches for waste classification. YOLOv4 and YOLOv5 models demonstrated faster real-time detection compared to SSD and EfficientNet architectures; however, their higher computational requirements make deployment on Raspberry Pi challenging. Such models are better suited for GPU-enabled environments in large-scale waste processing facilities.

2.1.2 Research Gaps of Existing Literature

Despite significant progress in AI-based waste classification and IoT-enabled waste management systems, several research gaps remain:

- Most deep learning models achieve high accuracy but require substantial optimization to operate efficiently on embedded and edge devices [1][3].
- IoT integration improves automation; however, limited processing power and memory of edge hardware restrict the deployment of complex AI models [2][6].
- Although real-time waste classification is feasible using lightweight CNNs, there is an inherent trade-off between classification accuracy and inference speed [3][5].
- Many existing systems provide limited or no support for wet biodegradable waste, reducing their effectiveness in real-world waste management scenarios [2][6].

2.1.3 Detailed Problem Analysis

Analysis of existing literature reveals that current smart waste segregation systems face several practical challenges when deployed outside controlled laboratory environments. Misclassification frequently occurs due to environmental factors such as poor lighting conditions, dirt accumulation on waste items, and overlapping or partially occluded objects [5]. These factors significantly degrade model performance in real-world settings.

Furthermore, many existing systems are limited to two or three waste categories, whereas practical applications demand segregation into at least four categories, including plastic, paper, and metal. Hardware limitations of IoT devices, such as constrained memory and processing capability, further restrict the use of advanced deep learning models on edge platforms [1][3].

Additionally, mechanical reliability and system scalability remain underexplored in many studies. These limitations collectively hinder the adoption of existing solutions for large-scale or institutional deployments.

2.1.4 Survey of Tools and Technologies Used

Based on the literature, the following tools and technologies are commonly employed in smart waste segregation systems:

- **Software Frameworks:** TensorFlow, Keras, PyTorch
- **AI Models:** AlexNet, MobileNet, InceptionResNet, VGG19
- **Hardware Platforms:** Raspberry Pi 4, Pi Camera modules, ultrasonic sensors, servo motors
- **IoT Communication Protocols:** MQTT, HTTP
- **Supporting Tools:** Python for system implementation and OpenCV for image preprocessing

These tools enable the development of end-to-end AI-based waste management systems integrating vision, computation, and mechanical actuation.

2.1.5 Summary

The literature indicates that AI-based waste classification combined with IoT infrastructure offers a promising approach to automated waste segregation. However, existing systems often suffer from limitations related to edge device constraints, limited waste categories, and real-world deployment challenges. The proposed SORT E-BIN system builds upon prior research by focusing on efficient edge deployment, multi-category waste classification, and tight integration of AI with mechanical sorting and IoT-based monitoring. This approach aims to bridge the gap between experimental prototypes and practical, deployable smart waste management solutions.

2.2 Software Requirement Specification

2.2.1 Introduction

2.2.1.1 Purpose

The purpose of SORT-E-BIN is to automate the process of waste segregation using deep learning-based image classification integrated with IoT-enabled hardware. The system eliminates manual effort, improves segregation accuracy, and provides timely notifications about bin fill levels. This contributes to:

- Improved recycling efficiency.
- Reduction of environmental hazards.
- Integration with smart city waste management infrastructures.

2.2.1.2 Intended Audience and Reading Suggestions

The primary audience of this Software Requirement Specification (SRS) document includes stakeholders from both technical and non-technical backgrounds who are directly or indirectly associated with the SORT-E-BIN project. This document is intended to serve as a reference for different categories of users. The municipal authorities and city administrators represent one of the most important groups of readers, as they are the potential adopters of such a system in smart city initiatives. For them, this document provides insights into how the system contributes to improved waste management efficiency, reduced manual effort, and environmental sustainability.

Another intended audience is researchers and academicians, particularly those working in the fields of Artificial Intelligence (AI), Internet of Things (IoT), and smart infrastructure. For them, this document highlights the integration of deep learning techniques with IoT-based hardware, giving them an understanding of how theoretical concepts can be implemented in real-world systems. This may inspire further extensions or optimizations, such as

deploying advanced neural networks, experimenting with transfer learning, or introducing explainable AI in waste classification.

Additionally, the document is meant for developers, system engineers, and designers who will be responsible for building, testing, and maintaining the system. For this group, the SRS provides technical specifications, interface requirements, and non-functional requirements, serving as a blueprint during the development phase. The end-users, including households, institutions, and communities, also form part of the audience. While they may not require in-depth technical details, this document ensures that they understand how the system operates, what features they can access, and how it improves their interaction with waste disposal processes.

2.1.1.1 Project Scope

The idea of SORT-E-BIN has been conceptualized to overcome the essential problem of waste segregation with the aid of image classification using AI and automation via IoT. The proposed scope of this particular project goes beyond an ordinary classification process and targets the design of an entirely different smart waste management system, which will be economical, extendable, and feasible. In essence, this segregation system identifies different types of trash into four bins, namely metal, paper, and plastic. The proposed SORT-E-BIN will ensure that each item goes into the respective trash bin, thereby removing the need for segregation by the user.

From a functional standpoint, the project consists of various modules that are interconnected. These modules include the acquisition of the image by the use of a Pi Camera, the classification process that involves the use of a deep learning model on the Raspberry Pi, the segregation process that involves the use of a servo motor to control the bins, as well as the use of ultrasonic sensors for the fill level assessment of the bins. In this project, there is the use of the LCD display, which provides the users with information regarding the detected category as well as the status of the bins. This enhances the creation of

awareness, as well as the instilling of confidence among the users due to the transparency of the operations.

The project not only aims to cater to small-scale utilization but has been developed for scalability. Although the initial implementation target may include the domain of household, university, or small-scale institutions, the system has been developed for expansion into municipal-scale smart city initiatives. The integration of cloud services may help in allowing a central monitoring of various bins in different locations, which can further help authorities in planning waste collection routes in order to cut costs and minimize environmentally adverse effects. The project's modular approach also enables easy implementation of extensions in the project, such as inclusion of additional types of waste, implementation of solar-modeled hardware, or control through a mobile app.

To conclude, the intended realm of application for the SORT-E-BIN system is as an end-to-end automated waste sorting system that not only has the functionality of intelligent sorting, but also makes way for more efficient monitoring capabilities through connected IoT technology. This helps fill the gap between laboratory-developed systems and their applicability in reality.

2.2.1 Overall Description

2.2.1.1 Product Perspective

SORT-E-BIN is designed as a standalone embedded system that combines machine learning algorithms with IoT-enabled hardware. Unlike conventional bins that rely on the user to decide where to dispose of waste, this system performs intelligent classification and segregation automatically. The system consists of four main layers:

- **Input Layer:** A Pi Camera that captures images of the waste item as soon as it is placed inside the bin.
- **Processing Layer:** A CNN-based classification model deployed on Raspberry Pi, responsible for analyzing the captured image and predicting the waste

category.

- **Actuation Layer:** Servo motors control the movement of the bin's lid or flap, ensuring waste is deposited into the correct compartment.

The system is modular in nature, meaning additional sensors, categories, or cloud-based services can be integrated in future versions. While SORT-E-BIN can operate independently at the household level, it also has the potential to be scaled up for deployment in universities, corporate campuses, and municipal smart cities.

2.2.1.3 Product Features

- **Automated Waste Classification:**

The system uses a trained Convolutional Neural Network (CNN) to automatically classify waste into four categories—plastic, paper, metal—in real time, ensuring fast and accurate decision-making at the point of disposal.

- **Automated Physical Segregation:**

A servo motor-driven mechanical mechanism directs the classified waste into its corresponding compartment, eliminating the need for manual sorting and preventing cross-contamination of waste streams.

- **Real-Time Bin Monitoring:**

Ultrasonic sensors are installed in each compartment to continuously monitor fill levels, enabling timely alerts and preventing bin overflow.

- **Edge-Based Processing:**

The waste classification model is deployed on an edge device (Raspberry Pi), allowing real-time inference with minimal latency and reduced dependency on cloud connectivity.

- **Scalability and Future Integration:**

The modular system design supports future expansion to additional waste categories, integration with IoT dashboards, and data analytics for institutional or municipal-level waste management.

2.2.2 External Interface Requirement

2.2.2.1 User Interfaces

The user interface is designed to be simple and intuitive so that individuals of all age groups can interact with the system without training. The main interface components are:

- **LCD Display:** Provides instant feedback by showing the identified waste category.
- **Dashboard (Future Scope):** A mobile or web-based application for municipal authorities to monitor fill levels across multiple bins.

2.2.2.2 Hardware Interfaces

The hardware components form the backbone of the system. Their interfaces are summarized as follows:

- **Raspberry Pi 4B:** Serves as the central controller, interfacing with all hardware components.
- **Pi Camera:** Captures images and sends them to Raspberry Pi for processing.
- **Servo Motors:** Controlled by Raspberry Pi GPIO pins to rotate or shift bin compartments.
- **Power Supply:** Provides stable voltage and current to Raspberry Pi and connected peripherals.

2.2.2.3 Software Interfaces

The software environment ensures smooth communication between hardware and AI models. It includes:

- **Operating System:** Raspbian OS for Raspberry Pi.
- **Libraries & Frameworks:** TensorFlow/Keras for CNN inference,

OpenCV for image preprocessing, NumPy for data handling.

- Programming Languages: Python (core implementation), with optional C/C++ for hardware-level control.
- IoT Protocols: MQTT or HTTP APIs for transmitting bin status data to remote dashboard.

2.2.3 Other Non-functional Requirements

2.2.3.1 Performance Requirement

The system must perform with a balance of accuracy, speed, and efficiency:

- Minimum classification accuracy of 85% across all categories.
- Response time of under 2 seconds per classification.
- Capable of handling 30–40 items per minute in public settings.
- Must function effectively even in low-light conditions or with partially damaged waste items.

2.2.3.2 Safety Requirements

As SORT-E-BIN involves electrical and mechanical components, safety considerations are critical:

- All electronic components are housed in a moisture-resistant casing to prevent short circuits.
- Mechanical moving parts (servo motors) are enclosed to prevent accidental injury.
- Power supply is equipped with fuses and surge protection.
- Special focus on handling wet waste without damaging sensors or

electronics.

2.2.3.3 Security Requirements

Since the system may be connected to IoT networks, data security and access control are essential:

- All data transmissions (bin fill levels, alerts) should be encrypted using secure protocols (e.g., SSL/TLS).
- Remote dashboards must have role-based access control to prevent unauthorized users from modifying or accessing sensitive data.
- The system should maintain error logs and allow manual override in case of software malfunction.

2.3 Cost Analysis

The cost analysis of the SORT-E-BIN system evaluates the financial feasibility of implementing the proposed solution. Since the project aims to be low-cost and scalable, the selection of hardware and software has been made with affordability and accessibility in mind, without compromising performance.

<u>Component</u>	<u>Quantity</u>	<u>Estimated Cost (₹)</u>	<u>Description</u>
Raspberry Pi 4B (4GB)	1	7000	Central processing unit for CNN inference and IoT control.
Pi Camera Module	1	1000	Captures images of waste items for classification.
Servo Motors (SG90/MG996R)	2-3	1500	Drive the rotating disc for segregation.
Rotating Disc (Acrylic/Metal)	1	1000	Acts as a base to direct waste towards the correct bin compartment.
LCD Display (16x2 or TFT)	1	1000	Provides feedback on waste category and bin status.
Power Supply & Adapters	1 set	800	Regulated power for Raspberry Pi and peripherals.
Miscellaneous (Cables, Connectors, Enclosure)	-	1200	Includes Wiring, protective casing and screws

Framework	4	3000	Hold the motors and camera
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Table 2.3 Cost Break-up

Total Estimated Prototype Cost: ~ **₹16,500-18,000**

2.4 Risk Analysis

Technical Risks

- Failure due to friction and other mechanical wear and tear of a servo motor.
- The disc might get stuck at times or misaligned while in operation.
- Raspberry Pi overheating or performance bottlenecks under heavy load.
- Poor light, occlusion, or contamination leading to misclassification of the waste.
- Ultrasonic sensors that provide faulty readings once covered with filth or wet waste.

Operational Risks

- Power supply interruptions or voltage fluctuations.
- Items being put into the receptacle by users in multiples/mixed and confusing the classifier.
- Difficulty in maintaining/calibrating sensors and motors over time.

External Risks

- Limited system throughput in public places during peak usage.

External Risks

- Environmental exposure to dust, moisture, and humidity destroys the electronics.
- Security vulnerabilities in IoT communication: data can be intercepted or even hacked.
- Financial constraints in scaling up for wide area deployment.
- Lack of awareness from the users or reluctance to adapt with automated bins.

CHAPTER 3 | METHODOLOGY ADOPTED

3.1 Investigative Techniques:

For the development of Sort E-bin, an AI-powered smart dustbin, a combination of **literature survey, experimental prototyping, and simulation-based investigation** has been adopted. The project required understanding both the **waste classification problem** (computer vision + machine learning) and the **mechanical actuation system** (servo-controlled disc mechanism).

a) Literature Review

- Reviewed existing research on **waste classification using CNN architectures** (e.g., InceptionResNetV2, VGG19, MobileNet).
- Studied prior **IoT-enabled waste segregation systems** (TrashBot, AIoT-based trash classification) to identify limitations (e.g., low accuracy, poor real-time integration).
- Identified the gap: most systems are either **software-only (image datasets)** or **hardware-only (sensor-based)**. Very few combine **AI + automation**

b) Dataset & Model Investigation

- Evaluated **public datasets (TrashNet, extended datasets)** for four categories: Plastic, Paper, metal.
- Performed **comparative training** with multiple CNNs (InceptionResNetV2, AlexNet, MobileNet, VGG19) and analyzed **accuracy, precision, recall, and F1-score**.
- Justified final choice of **InceptionResNetV2** based on superior accuracy for multi-class waste classification

c) Experimental Prototyping

- Designed a **disc-based segregation mechanism**.

- Investigated **servo torque requirements**, IR sensor sensitivity, and camera module positioning to capture images clearly before classification.
- Prototyped using **Raspberry Pi 4** as the edge device, testing computational load and inference time.

d) Simulation & Testing

- Simulated **servo actuation angles** (0° = Paper, 120° = Plastic, 240° = Paper) to verify correct disposal.
- Conducted **confusion matrix analysis** for each model to validate classification accuracy.
- Tested multiple thresholds (τ) for confidence filtering to handle ambiguous cases.

Justification:

The investigative approach (review $\rightarrow$ dataset/model evaluation $\rightarrow$ hardware prototyping $\rightarrow$ simulation/testing) ensures that the solution is evidence-based, feasible, and replicable. It balances AI model investigation with hardware integration testing, which is essential for real-world deployment

3.2 Proposed Solution

The proposed solution, that is Sort E-bin, is a low-cost, **AI-powered waste segregation bin** that sorts garbage into four distinct categories: Plastic, Paper, Metal

Core Workflow

1. **Sensing (Camera):**

- Detects whether the disc is empty or occupied.
- Ensures the system only activates when garbage is present.

2. **Vision Capture & Classification:**

- Camera captures the item image.
- Image will be pre-processed (resize, normalization).
- Inception-ResNet-V2 (and other benchmarked models) will predict the waste category.
- If confidence < threshold (τ), classify into “Others” for safety.

3. **Decision & Actuation:**

- Raspberry Pi maps the predicted class to a **servo motor rotation angle**:
 - $0^\circ \rightarrow$ Paper
 - $120^\circ \rightarrow$ Plastic
 - $240^\circ \rightarrow$ Metal
- A secondary servo motor tilts the disc to dispose the item into the selected bin.
- Disc resets back to 0° .

4. **Feedback & Logging:**

- User receives real-time status (“Plastic Detected”, “Disposed Successfully”).
- Calibration updates (angles, tilt duration) possible through admin input

Advantages over Existing Solutions:

- **Multi-category sorting (3 types)** vs. conventional 2-type smart bins.
- **Confidence-based fallback** prevents mis-classification.
- **Low-cost hardware** (Raspberry Pi + servo motors) makes it scalable for institutions.

- **Modular architecture** allows easy addition of new categories in the future.

3.3 Work Breakdown Structure

To ensure systematic development and integration of both hardware and software components, we divided the project into modules and sub-tasks following a Work Breakdown Structure (WBS). This structure allows for parallel progress, clear responsibility allocation, and step-by-step validation of each part before integration into the final prototype.

1. **Data Acquisition and Model Training**

A dataset of waste images was collected and preprocessed. Various models (InceptionResNetV2, AlexNet, MobileNet, VGG19) were trained, and their accuracy was compared.

2. **Hardware Sensing and Control**

- Integration of the Camera for object presence detection.
- Camera setup for image capture.
- Servo motors are configured for rotation and tilting.

3. **Software Integration**

- Deployment of the final trained model on Raspberry Pi.
- Python scripts for end-to-end coordination between sensors, model, and motors.

4. **Prototype Development and Testing**

- Assembly of the disc and compartments.
- Full integration of software and hardware.
- Testing under different scenarios (single item, mixed waste, repeated cycles).

S. No.	Module	Sub-Tasks	Deliverables
1	Data Collection & Training	- Collect dataset (TrashNet + custom images) - Preprocess and label data - Train models (InceptionResNetV2, AlexNet, MobileNet, VGG19)	Trained ML models with accuracy comparison
2	Hardware Setup	- Install an IR sensor for object detection - Integrate the camera module for image capture - Connect and configure Servo Motors (rotation & tilt)	Functional hardware prototype with sensors and motors
3	Software Integration	- Deploy final trained model on Raspberry Pi 4 - Write Python scripts for motor control and sensor interfacing - Display integration for category output	End-to-end working software integrated with hardware
4	Prototype Development	- Assemble rotating disc and compartments - Calibrate servo angles (0°, 90°, 180°, 270°) - Align bins with categories	Assembled working prototype
5	Testing & Evaluation	- Test system with different waste items - Record classification accuracy and motor response - Debug and optimize performance	Final performance report and validated prototype

Table 3.3 Work Breakdown Structure

3.4 Tools and Technology

Hardware

- Raspberry Pi 4 Model B (4GB RAM): Central controller.
- IR Sensor: Presence detection.
- Camera Module (Raspberry Pi Camera v2 / USB Camera): Image capture.
- Servo Motors (2x): Disc rotation + tilt disposal.
- Circular Disc: Garbage placement platform.
- Bins (x4): For segregated categories.

Software / Tools

- Programming Languages: Python 3.9 (for ML + hardware integration).
- Deep Learning Frameworks: TensorFlow, Keras.
- Supporting Libraries: OpenCV (image preprocessing), NumPy, Pandas, Matplotlib (visualizations).

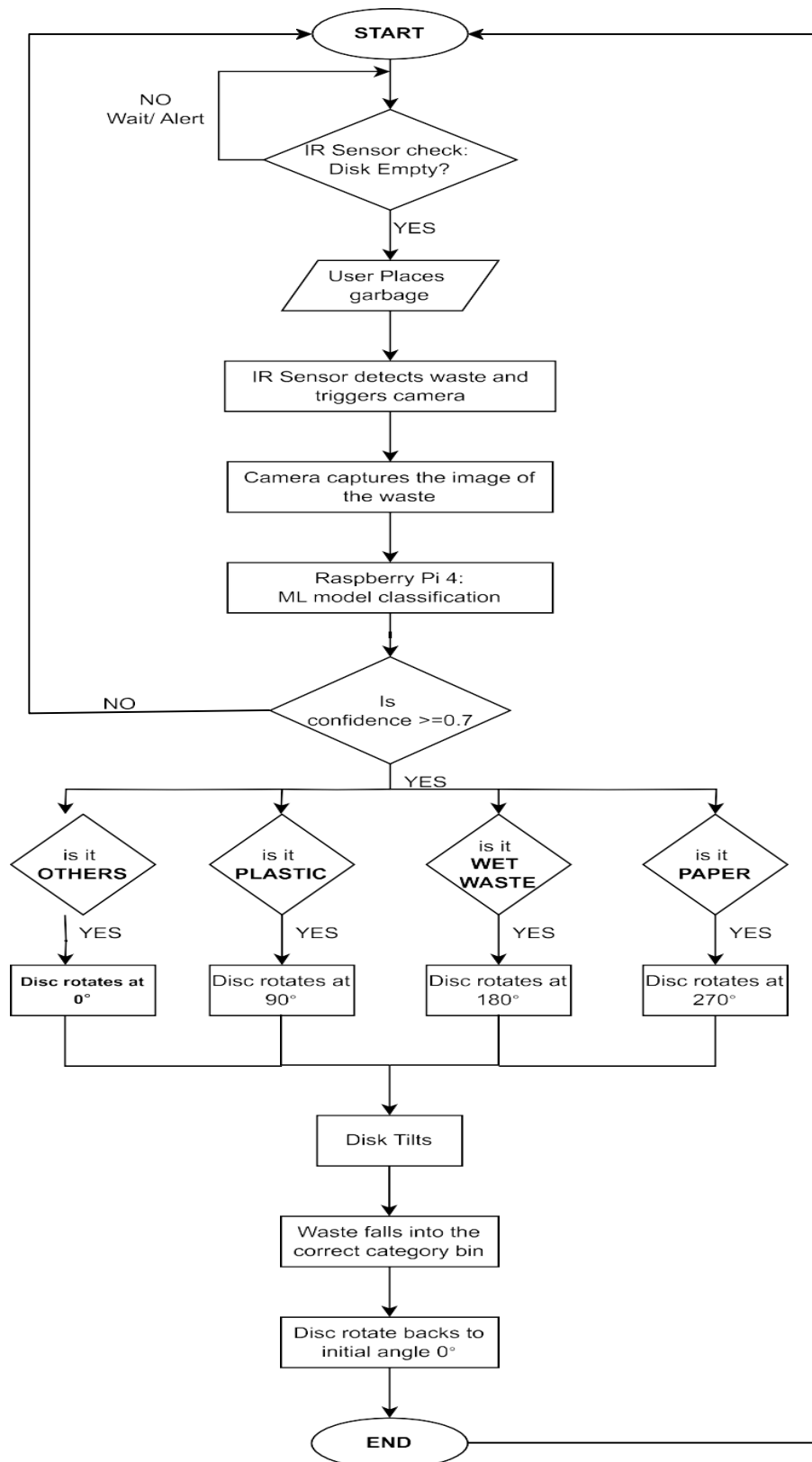
- OS: Raspbian OS (on Raspberry Pi).
- Version Control: Git/GitHub for code management.

Model Training Environment

- Google Colab / Jupyter Notebook with GPU acceleration.
- Pretrained CNN architectures: Inception-ResNet-V2, VGG-19, MobileNet, AlexNet.
- Dataset: TrashNet + custom images (resized, cleaned, augmented).

CHAPTER 4 | DESIGN SPECIFICATION

4.1 System Architecture



This flowchart illustrates the complete operational workflow of the SortEbin Smart Bin, starting from waste detection to final segregation into the correct category bin. Each step is logically sequenced and ensures reliable automation.

Start

The process begins when the system is powered on and is in idle/ready state.

Check if Disc is Empty (IR Sensor)

The IR sensor continuously checks if the circular disc is empty.

If disc is not empty → Wait/Alert.

If empty → continue.

User Places Garbage

Once the disc is detected empty, the user places an item (waste) on the disc.

Waste Detection (IR Sensor Trigger)

The IR sensor detects the presence of waste and triggers the camera.

Image Capture (Camera)

The camera captures an image of the waste placed on the disc.

Waste Classification (Raspberry Pi + ML Model)

The image is processed by the Raspberry Pi 4 running the selected deep learning model (Inception-ResNet-V2 / MobileNet etc.).

The model predicts the waste category and outputs a confidence score.

Confidence Threshold Check (≥ 0.7)

If the prediction confidence is less than 0.7, the waste is classified into “Others” for safety.

If confidence is ≥ 0.7 , the system checks for the predicted category.

Category Classification

The waste is classified into one of four categories:

Others $\rightarrow$ Disc rotates to 0°

Plastic $\rightarrow$ Disc rotates to 90°

Wet Waste $\rightarrow$ Disc rotates to 180°

Paper $\rightarrow$ Disc rotates to 270°

Disc Tilt Operation

After rotation, the disc tilts downward to drop the item into the correct bin.

Waste Disposal

The waste falls into the respective segregated bin.

Reset Position

Once disposal is complete, the disc rotates back to its initial angle (0°) for the next cycle.

End

The cycle ends, and the system returns to the idle state, waiting for the next input.

4.2 Design Level Diagrams

4.2.1 Level-0 Context DFD

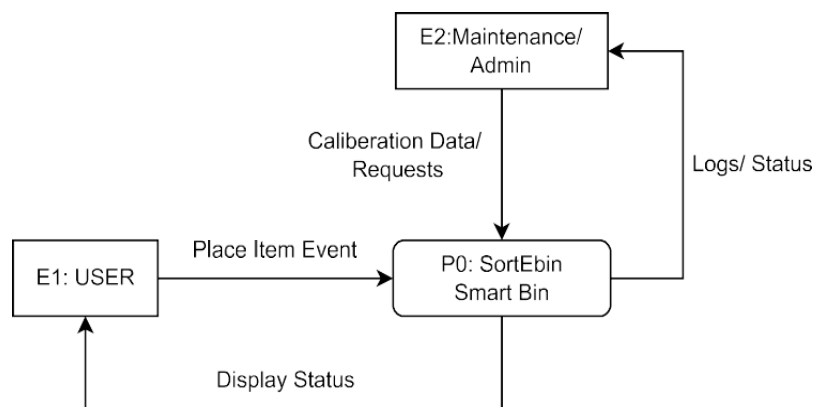


Figure 4.2.1: Level-0 Context DFD

The diagram illustrates the high-level interaction between the SortEbin Smart Bin system and its external entities. At this level, the entire system is represented as a single process (P0), and only its input/output relationships with users and administrators are shown.

Entities and Process

E1: User (External Entity)

Represents the person who interacts with the bin.

Action: Places garbage on the disc (logical event).

Receives: Status feedback from the bin such as “Plastic Detected”, “Disposed Successfully”, or error messages.

E2: Maintenance/Admin (External Entity)

Represents the maintenance personnel responsible for calibration, monitoring, and servicing.

P0: SortEbin Smart Bin (Process)

Central system that manages the sensing, classification, actuation, and feedback operations.

Interfaces with both the User (E1) and the Admin (E2) to ensure smooth functioning and proper waste segregation.

Data Flows

E1 → P0: PlaceItemEvent

Logical trigger when the user places an item, detected via the IR sensor.

P0 → E1: DisplayStatus

Feedback to the user, e.g., detected waste category or operational errors.

E2 → P0: Calibration Data / Requests

Input from admin to update calibration parameters such as servo angles, tilt duration, or sensitivity.

P0 → E2: Logs / Status

Outputs operational data like cycle logs, error reports, and usage analytics for monitoring.

4.2.2 Level-1 DFD

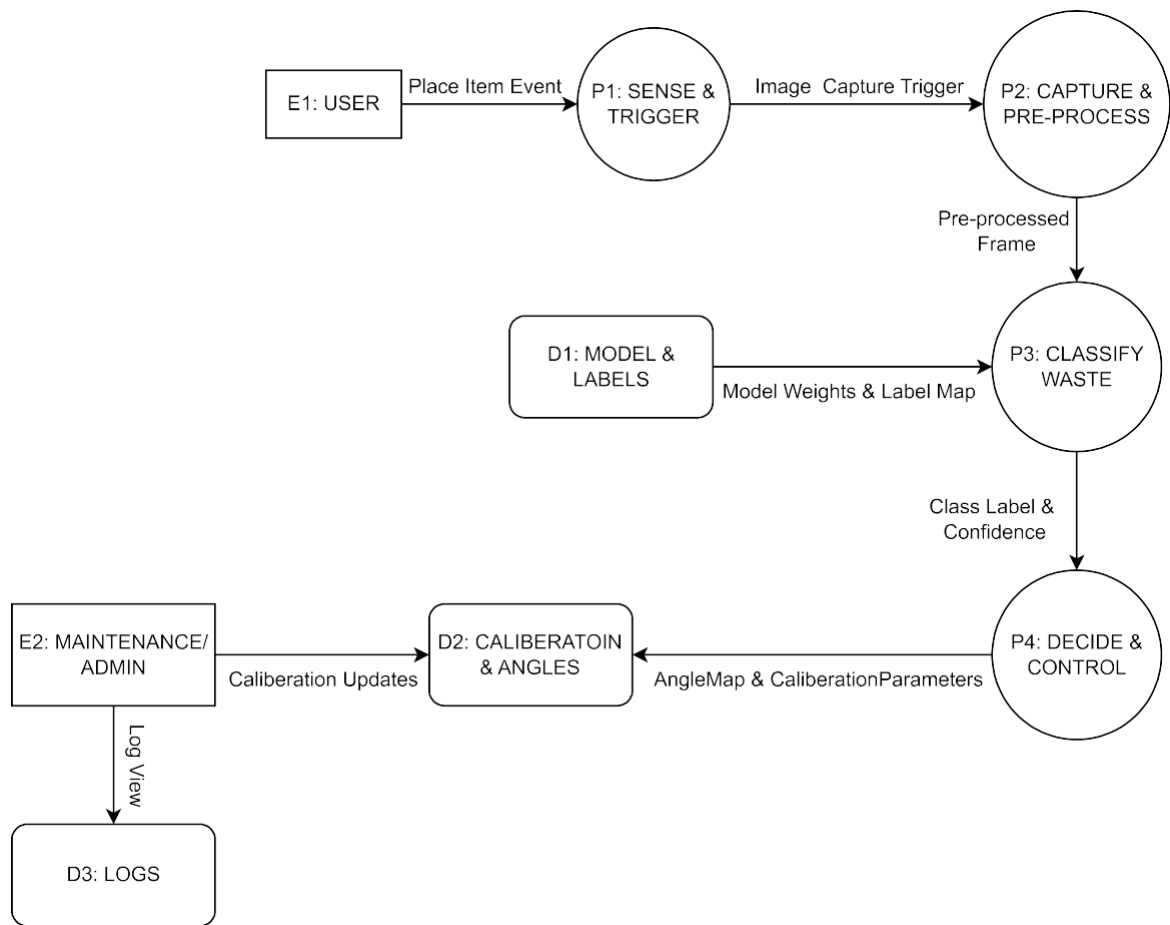


Figure 4.2.2: Level-1 DFD of SortEbin (Sensing → Vision → Decision → Actuation)

Level-1 DFD

The Level-1 DFD expands the single process (P0) of the Level-0 diagram into multiple subprocesses. It illustrates the detailed workflow of sensing, vision, decision-making, actuation, and logging. This provides a clear understanding of how data flows through each stage of the system.

Entities, Processes, and Data Stores

External Entities

E1: User

Places garbage on the disc.

Triggers the sensing process via PlaceItemEvent.

E2: Maintenance/Admin

Provides calibration updates (servo angles, tilt timing).

Can access logs for monitoring system performance.

Processes

P1: Sense & Trigger

Uses IR sensor to check disc presence.

Generates a CaptureTrigger for the camera when an item is detected.

P2: Capture & Preprocess

Camera captures an image of the waste.

Image is resized and normalized before classification.

P3: Classify Waste

Processes the preprocessed image using ML model.

Outputs a predicted Class Label & Confidence.

Uses reference data from D1: Model & Labels.

P4: Decide & Control

Maps predicted class into specific servo angle.

Sends Servo Command to actuators.

Reads and writes calibration data from D2: Calibration & Angles.

P5: Feedback & Logging (not shown at top but implied in D3 linkage)

Updates display with detected class/status.

Stores cycle logs into D3: Logs.

Data Stores

D1: Model & Labels

Stores ML model weights and label mappings.

Accessed by P3 for classification.

D2: Calibration & Angles

Holds servo rotation angles, tilt degree, and timing parameters.

Updated by Admin (E2).

Read by P4 for decision mapping.

D3: Logs

Stores cycle logs, predictions, confidence levels, and errors.

Can be accessed by Admin for review.

4.2.3 Level-2 DFD for P3: Classify Waste

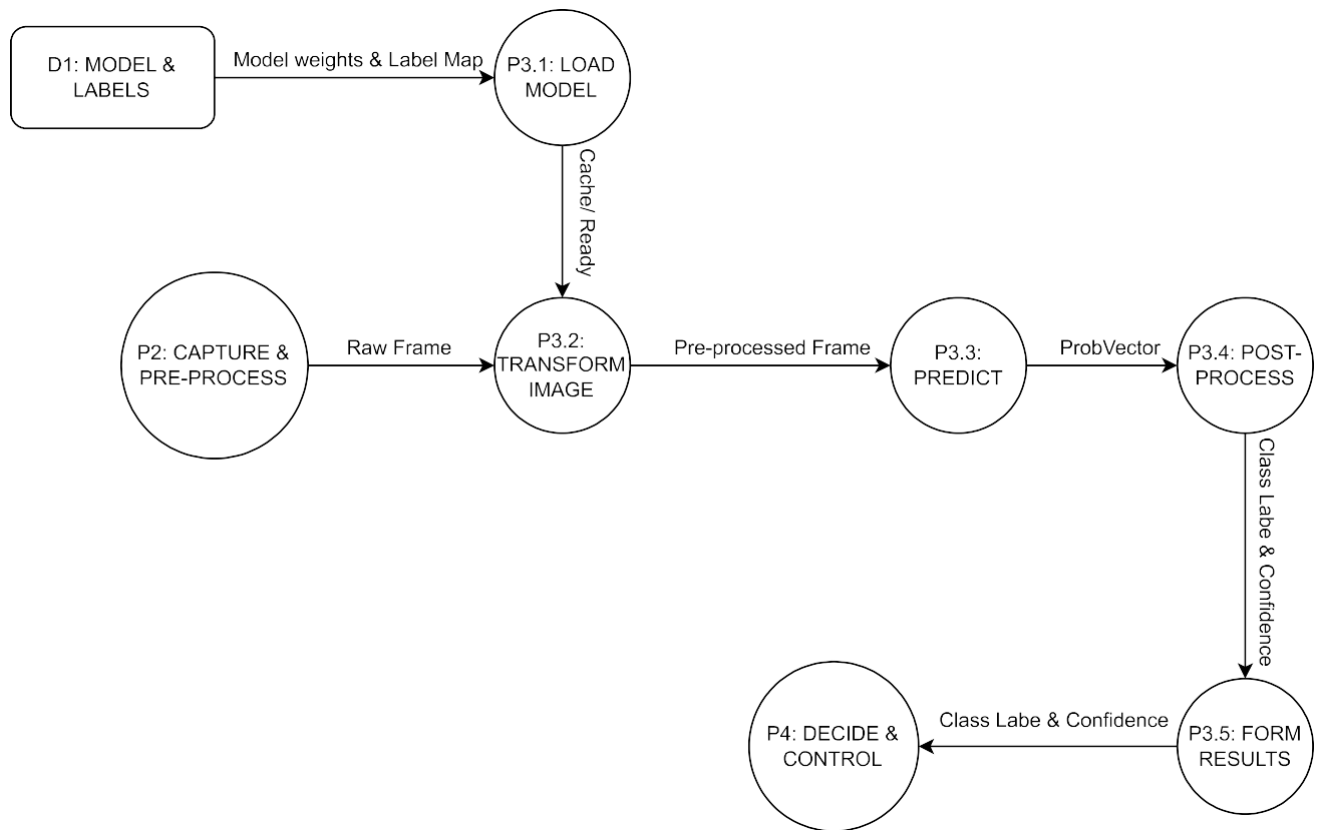


Figure 4.2.3: Level-2 DFD for P3 (Classify Waste)

Level-2 DFD (P3: Classify Waste)

This diagram expands the single process P3: Classify Waste (from Level-1 DFD) into detailed subprocesses. It explains how raw image data flows through different ML stages until a final class label and confidence score are produced.

Processes in the Diagram

P3.1 Load Model

At system startup (or when cached), the trained ML model and its label map are loaded from D1: Model & Labels.

This ensures that the classification pipeline always has the correct trained weights.

P3.2 Transform Image

Takes the Raw Frame from P2: Capture & Preprocess.

Performs resizing, cropping, normalization, and other image preprocessing.

Outputs a Pre-processed Frame suitable for ML inference.

P3.3 Predict

The preprocessed frame is passed through the loaded deep learning model (e.g., Inception-ResNet-V2).

The model produces a Probability Vector (ProbVector), which indicates the likelihood of the waste belonging to each category.

P3.4 Post-Process

Analyzes the probability vector.

Applies argmax (highest score) and a confidence threshold (τ) to ensure reliability.

If confidence $< \tau$, the item is classified as “Others” to avoid misclassification.

P3.5 Form Result

Converts the post-processed output into a final Class Label & Confidence Score.

This result is passed to P4: Decide & Control for actuation.

Data Stores & Flows

D1: Model & Labels → P3.1 Load Model

Supplies the trained ML weights and label mappings.

P2 Capture & Preprocess → P3.2 Transform Image

Provides raw frames for preprocessing.

P3.2 → P3.3 → P3.4 → P3.5

Sequential data flow: Preprocessed Frame → Probability Vector → Class Label & Confidence.

P3.5 → P4 Decide & Control

Sends final classification result for servo movement decisions.

4.2.4 Level-2 DFD for P4: Decide & Control

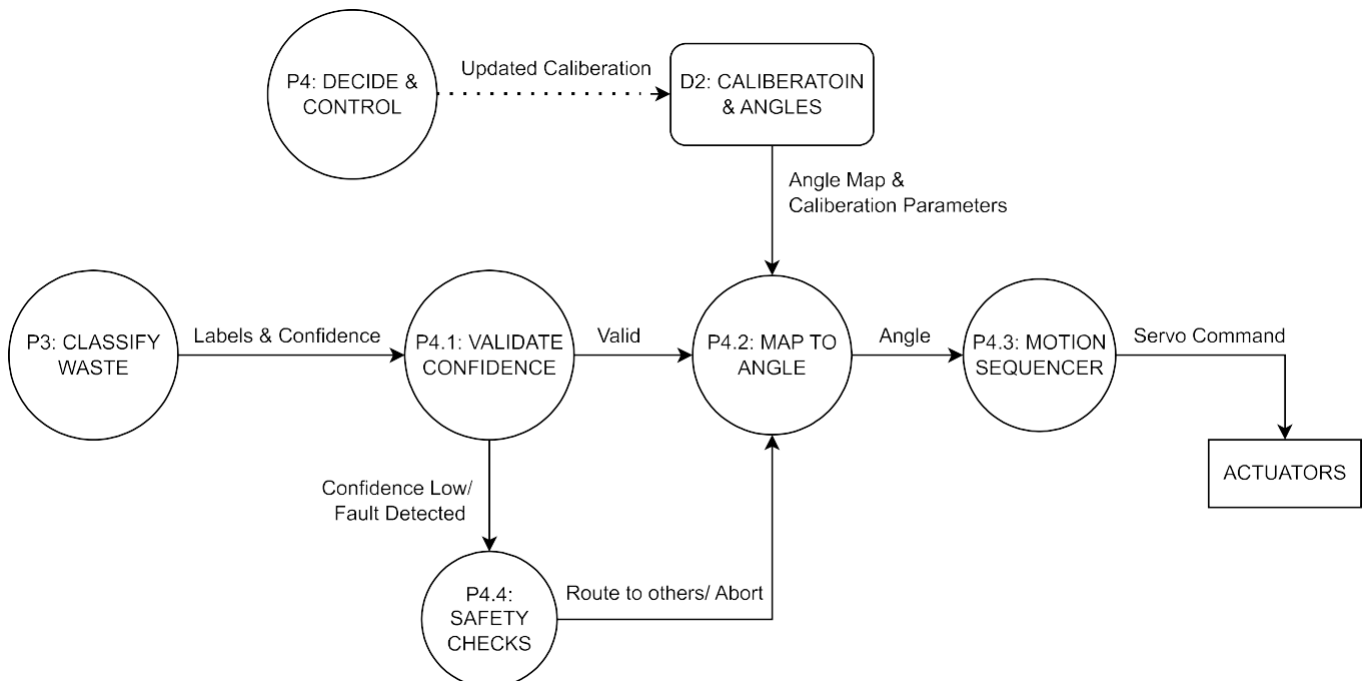


Figure 4.5: Level-2 DFD for P4 (Decide & Control)

Level-2 DFD (P4: Decide & Control)

This diagram expands P4: Decide & Control (from Level-1 DFD) into multiple subprocesses that handle confidence validation, mapping to servo angles, motion sequencing, and safety

checks. It shows how the system translates classification results into physical actions to sort waste.

Subprocesses in the Diagram

P4.1 Validate Confidence

Input: Class Label & Confidence from P3: Classify Waste.

Checks whether the classification confidence is above the threshold (τ).

If confidence is too low or a fault is detected → forward to P4.4 Safety Checks for handling.

If valid → continue to mapping stage.

P4.2 Map to Angle

Uses the class label to decide which angle the servo motor should rotate to.

Consults D2: Calibration & Angles (which stores angle map, tilt degree, and timing parameters).

Output: Angle for servo actuation.

P4.3 Motion Sequencer

Converts the mapped angle into servo commands for actuators.

Handles step-by-step motor actions: Rotate → Tilt → Return to initial position.

Ensures waste falls into the correct bin.

Output: Servo Command to actuators.

P4.4 Safety Checks

Triggered if:

Confidence is below threshold.

Faults are detected (e.g., IR sensor shows disc not empty).

Ensures the system avoids unsafe or wrong actions.

Routes item to “Others” bin or aborts action.

Data Stores

D2: Calibration & Angles

Provides servo calibration data (angle map, tilt angle, hold time).

Can also be updated by P4 Decide & Control when maintenance adjusts parameters.

Data Flows

CHAPTER 5 | IMPLEMENTATION AND EXPERIMENTAL RESULTS

5.1 Experimental Setup

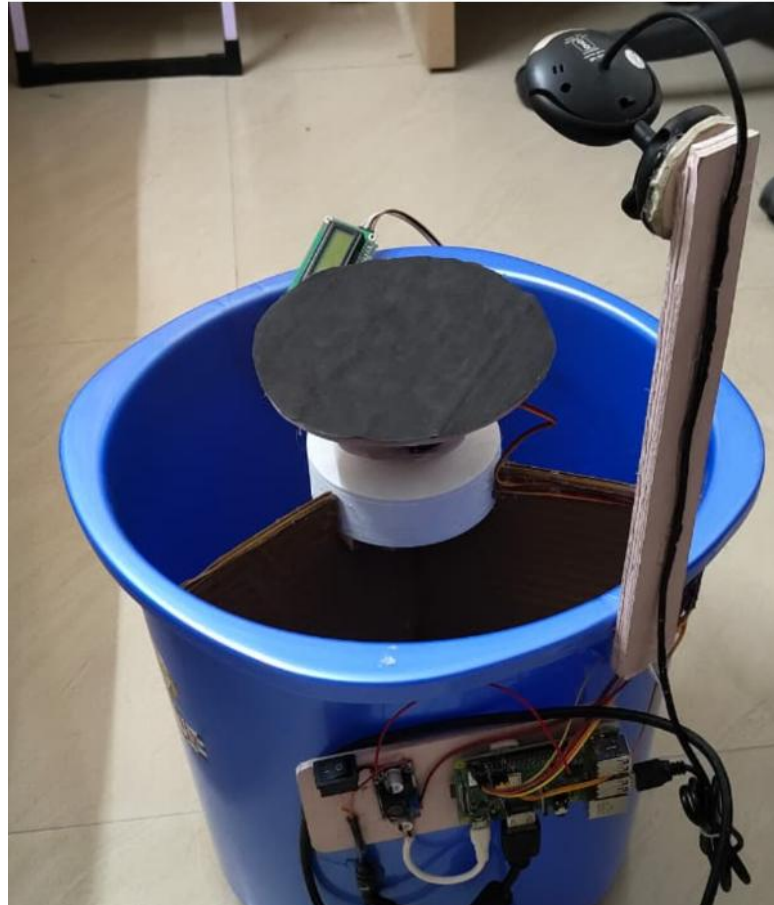


Figure 5.1(a): Hardware setup (Raspberry Pi, camera, motors)



Figure 5.1(b): LCD Setup (Disc is empty, so **No Waste** Detected)



Figure 5.1(c): Places garbage on the Disk
(plastic)



Figure 5.1(d): Waste segregation in action
(plastic)



Figure 5.1(e): LCD Display (**Plastic**
Detected)

5.2 Experimental Analysis

The experimental analysis focuses on understanding the quality of input data, the learning behavior of the model, and the effectiveness of the waste classification system under realistic conditions.

5.2.1 Data

(Data Sources/Data Cleaning/Data Pruning/ Feature Extraction Workflow)

The performance of any machine learning system largely depends on the quality and relevance of the data used. For the SORT E-BIN project, careful attention was given to data collection and preprocessing to ensure that the trained model performs reliably in real-world waste segregation scenarios.

- Data Sources

The dataset used for this project consists of images belonging to three recyclable waste categories: metal, paper, and plastic. The data was obtained from a combination of publicly available waste image datasets and custom images captured specifically for this project. Public datasets provided a wide variety of waste samples, while custom images were collected using a camera module to reflect real usage conditions such as crumpled paper, partially visible plastic wrappers, and reflective metal containers commonly found in campus environments. This combination helped in creating a dataset that is both diverse and practically relevant.

- Data Cleaning

Before training the model, the dataset was carefully examined to remove unsuitable samples. Images that were corrupted, duplicated, excessively blurred, or poorly framed were removed. In some cases, images where the waste object was not clearly visible or was heavily occluded were also excluded. This cleaning process helped reduce noise in the dataset and ensured that the model learned meaningful visual patterns rather than irrelevant background information.

- Data Pruning

Data pruning was performed to maintain a balanced representation of waste categories. Redundant images with very similar visual characteristics were selectively reduced, especially in classes with a higher number of samples. This step helped prevent bias toward any specific category and improved the model's ability to generalize during validation and testing. By pruning unnecessary data, training efficiency was also improved without compromising classification performance.

- **Feature Extraction Workflow**

Feature extraction in the SORT E-BIN system is handled automatically by the deep learning model. The InceptionResNetV2 architecture extracts visual features such as edges, textures, shapes, and material-specific patterns directly from the input images. Transfer learning was used by initializing the network with ImageNet-trained weights, allowing the model to reuse previously learned low-level features. During training, higher layers of the network were fine-tuned so that the extracted features became more relevant to waste classification. This approach significantly reduced training time while improving classification accuracy, even with a limited amount of domain-specific data.

5.2.2 Performance Parameters

(Accuracy Measures and Classification Metrics)

To evaluate the effectiveness of the proposed waste classification system, standard performance metrics used in multi-class classification problems were employed. These metrics provide a detailed understanding of model behavior across different waste categories.

- **Accuracy**

Accuracy represents the proportion of correctly classified waste samples out of the total number of samples evaluated. The proposed InceptionResNetV2-based model achieved an overall accuracy of approximately 84%, indicating that the system can reliably classify waste into metal, paper, and plastic categories under realistic conditions.

- Precision

Precision measures how often the model's predictions for a given class are correct. High precision values were observed for paper and plastic, which indicates that when the model predicts these categories, it is usually correct. A comparatively lower precision for metal was observed, mainly due to reflective surfaces that visually resemble certain plastic materials.

- Recall

Recall evaluates the ability of the model to correctly identify all samples belonging to a specific class. The model demonstrated strong recall for metal and paper waste, showing that most of the actual instances of these categories were successfully detected during testing.

- F1-Score

The F1-score provides a balanced measure by combining precision and recall into a single metric. The obtained F1-scores confirm that the model maintains a reasonable balance between false positives and false negatives across all waste categories, making it suitable for real-world deployment.

Classification Report:				
	precision	recall	f1-score	support
metal	0.71	0.94	0.81	456
paper	0.91	0.97	0.94	454
plastic	0.93	0.66	0.77	596
accuracy			0.84	1506
macro avg	0.85	0.86	0.84	1506
weighted avg	0.86	0.84	0.83	1506

Figure 5.2.2(a): Accuracy Measures and Classification Metrics

- Confusion Matrix Analysis

The confusion matrix offers a class-wise view of the model's predictions. Most values are concentrated along the diagonal, indicating correct classification. Some misclassification between metal and plastic was observed, which can be attributed to similar visual properties such as shine and reflectivity under varying lighting conditions. Overall, the confusion matrix confirms the robustness of the proposed classification approach.

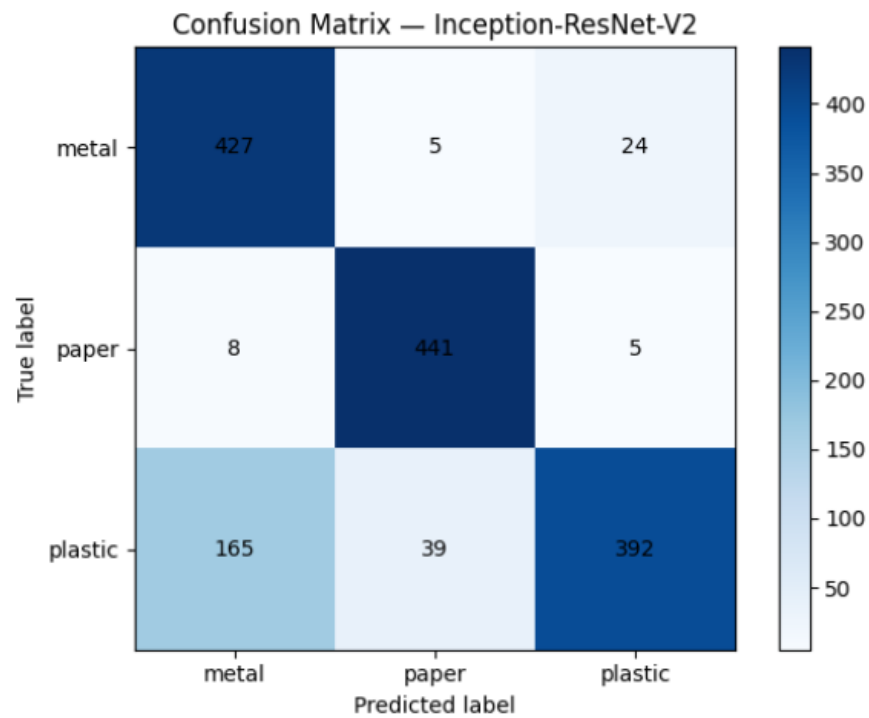


Figure 5.2.2(b): Confusion Matrix

5.3 Working of the project

5.3.1 Procedural Workflow

The SORT E-BIN system follows a structured procedural workflow that integrates real-time image processing, intelligent waste classification, and automated mechanical actuation. The workflow is designed to operate continuously and autonomously, ensuring reliable waste segregation with minimal user involvement.

The process begins when the system is powered on. During initialization, the Raspberry Pi configures all connected peripherals, including the USB camera,

servo motors, and LCD display. The trained waste classification model is loaded into memory, and the system enters an idle monitoring state.

Once initialized, the USB camera continuously captures live video frames of the waste inlet area. The system monitors each frame to detect the presence of an object placed on the base platform. If no object is detected, the system remains in standby mode without activating any mechanical components.

When a waste item is placed on the base platform, the camera captures the image in real time. The captured frame is forwarded to the preprocessing module, where it undergoes resizing and normalization. These preprocessing steps ensure that the image matches the expected input format and reduces variations caused by lighting conditions, shadows, or camera distance.

After preprocessing, the image is passed to the waste classification module running on the Raspberry Pi. Internally, the classification logic analyzes visual characteristics such as color distribution, brightness, and texture patterns to determine whether a valid waste object is present. If the image does not contain a detectable waste object, the system classifies it as “No Waste” and takes no further action.

If a valid waste object is detected, the classification module determines the waste category as plastic, paper, or metal. Along with the predicted label, a confidence value is generated to assess classification reliability. The predicted category is then passed to the decision-making module.

The decision-making module maps each waste category to a predefined mechanical action. Internally, specific angular positions are assigned to each bin compartment. Based on the predicted waste type, control signals are generated and sent to the servo motor controller through the Raspberry Pi’s GPIO interface.

The first servo motor rotates the disc mechanism to align the waste outlet with the appropriate bin compartment. Once alignment is achieved, the second servo motor is activated to push or tilt the disc, allowing the waste item to fall into the selected compartment. This two-step actuation ensures accurate and controlled disposal of waste.

Simultaneously, the detected waste category is displayed on the LCD screen, providing immediate visual feedback to the user. This feedback confirms successful detection and segregation, improving user trust and system transparency.

After disposal, both servo motors return to their default positions. The system then resets its internal state and resumes continuous monitoring for the next waste item. A short delay is maintained between successive detections to prevent repeated classification of the same object.

This closed-loop procedural workflow enables SORT E-BIN to operate efficiently in real-time environments, ensuring consistent waste classification, precise mechanical segregation, and continuous autonomous operation.

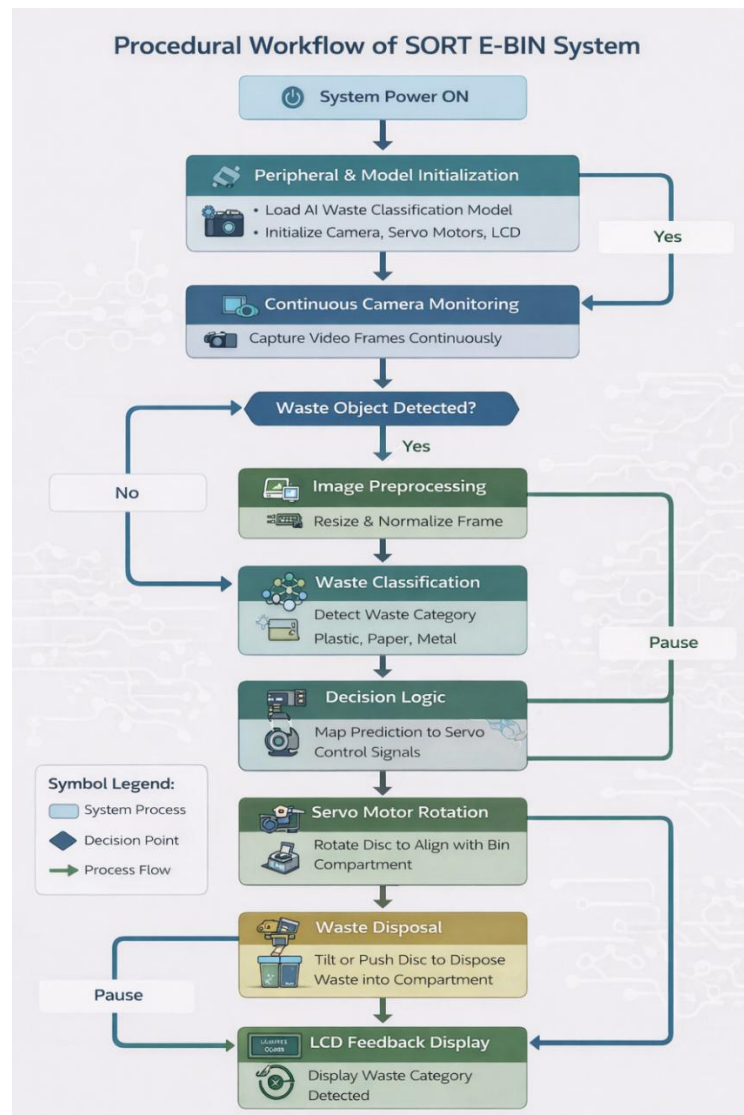


Figure 5.3.1: Workflow diagram

5.3.2 Algorithmic Approaches Used

SORT E-BIN system employs two primary algorithmic components: a waste classification algorithm and a decision-based actuation algorithm. These algorithms work together to convert visual input into physical segregation.

- **Algorithm 1: Waste Detection and Classification Algorithm**

The waste classification process is implemented as a lightweight inference pipeline optimised for embedded execution. Each camera frame is analysed to detect the presence of waste and classify it into predefined categories.

Instead of relying on cloud-based inference, the model runs locally on the Raspberry Pi, ensuring low latency and independence from network connectivity. Feature extraction is performed by analysing visual characteristics such as colour distribution and brightness patterns, which are effective indicators for distinguishing plastic, paper, and metal waste.

Pseudocode:

```
Capture image frame
Resize and normalize image
Analyze visual features
If object not detected:
    Return "No Waste"
Else:
    Determine waste category
Return predicted label and confidence
```

- **Algorithm 2: Decision and Actuation Control Algorithm**

Once the waste category is identified, a rule-based control algorithm determines the mechanical response. Each category is associated with a predefined servo motor angle. Pseudocode:

```
Input predicted waste label
If label is Plastic:
    Rotate selection servo to plastic angle
Else if label is Paper:
    Rotate selection servo to paper angle
Else if label is Metal:
    Rotate selection servo to metal angle
Activate push servo to dispose waste
Reset servos to default position
Display label on LCD
```

This deterministic approach ensures reliable and repeatable mechanical operation, even under continuous usage.

5.3.3 Project Deployment

The SORT E-BIN system is deployed as a fully edge-based intelligent waste segregation unit. The Raspberry Pi acts as the central processing and control hub, hosting the waste detection logic, motor control routines, and display interface. The trained classification logic is stored locally on the Raspberry Pi, enabling real-time inference without dependency on external servers. The USB camera provides continuous visual input, while servo motors are controlled using GPIO-based pulse-width modulation signals. An I2C-based LCD module provides real-time user feedback. The system power supply includes a DC–DC buck converter to regulate voltage levels and ensure stable operation of all components. The modular software design allows independent operation of image processing, decision-making, actuation, and display modules, simplifying maintenance and future upgrades.

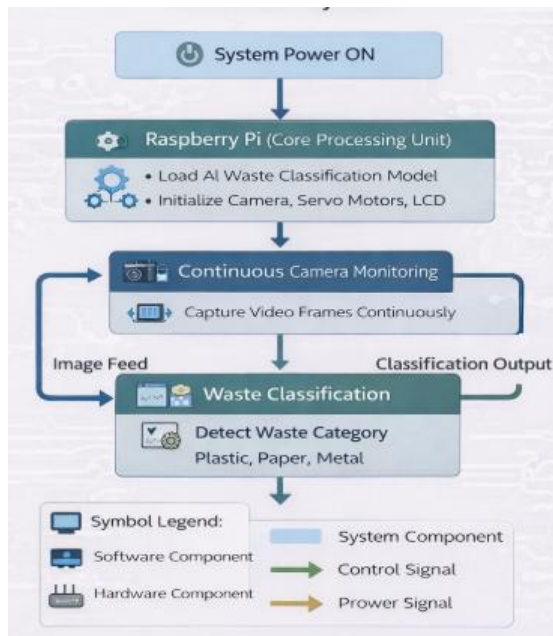


Figure 5.3.3(a): Component diagram

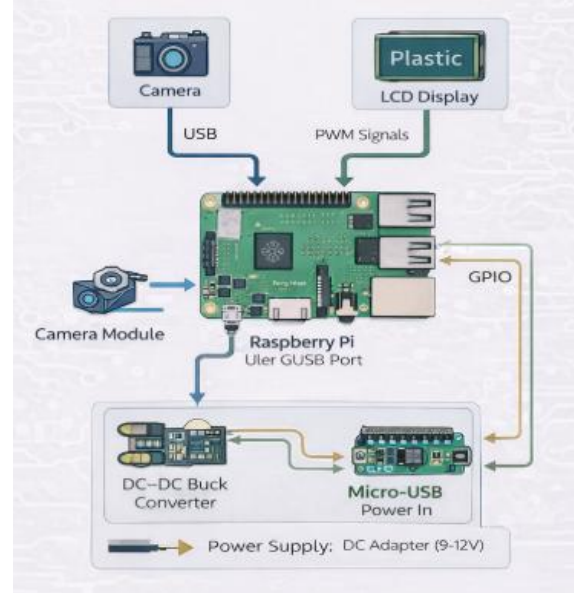


Figure 5.3.3(b): Deployment diagram

5.3.4 System Screenshots

Figure 5.3.4.1: Camera capturing waste object on base platform



Figure 5.3.4.2: LCD displaying detected waste category



Figure 5.3.4.3: Servo motor rotating disc mechanism

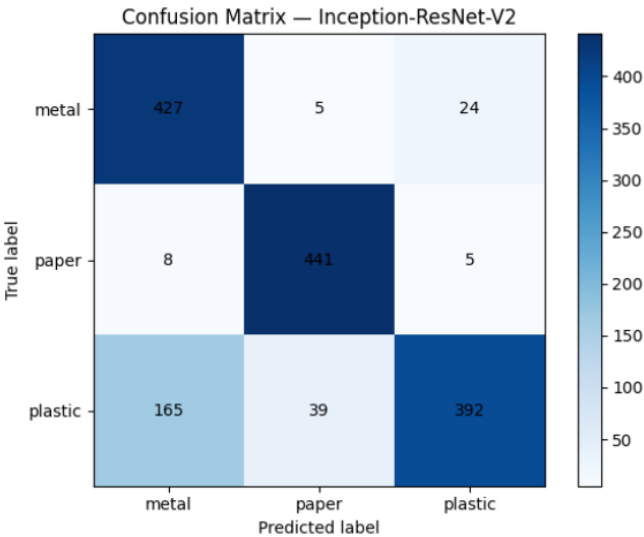


Figure 5.3.4.4: Confusion matrix of waste classification results

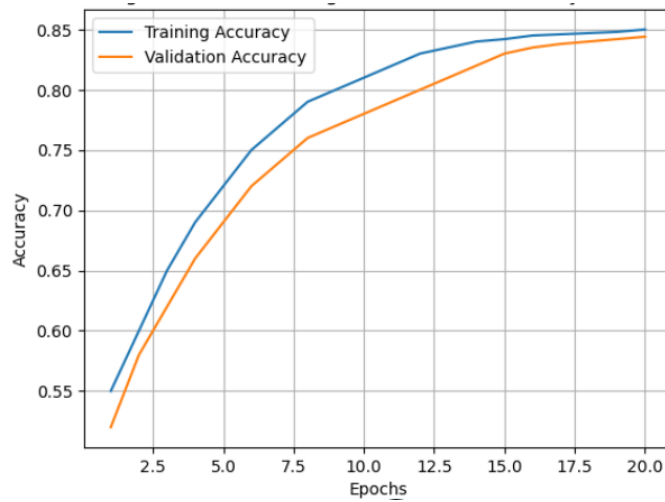


Figure 5.3.4.5(a): Accuracy and loss curves of the classification model

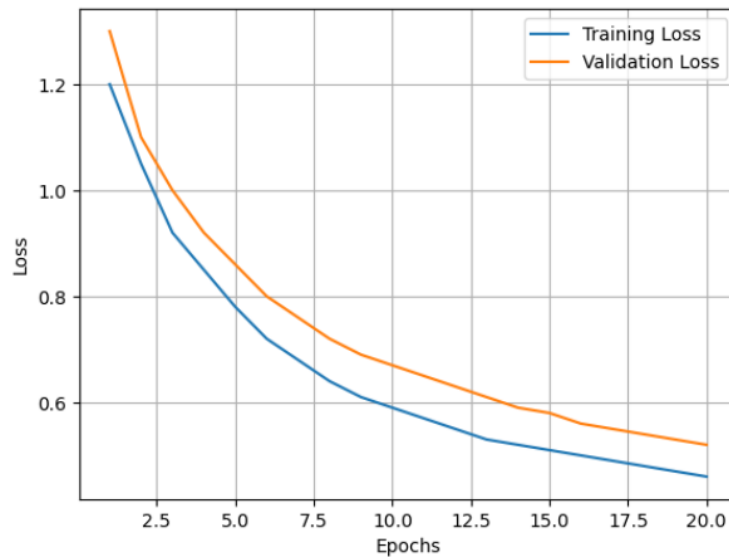


Figure 5.3.4.5(b): Accuracy and loss curves of the classification model

5.4 TESTING PROCESS

Testing is a critical phase in the development of the **AI-Powered Smart Waste Sorting System**, as it ensures that the implemented system functions correctly, efficiently, and reliably under both controlled and real-world conditions. The testing process was carried out to validate the performance of the software modules, hardware components, and their seamless integration. A structured and systematic testing methodology was adopted to evaluate system accuracy, responsiveness, and stability.

The testing phase focused on verifying that the system correctly identifies waste categories using the trained deep learning model, performs accurate mechanical segregation using servo motors, and provides appropriate user feedback through the display unit.

5.4.1 Test Plan

The test plan outlines the overall approach adopted to evaluate the functionality and performance of the proposed system. The primary objectives of the test plan are:

- To verify correct operation of individual software and hardware modules
- To validate the accuracy of AI-based waste classification
- To ensure proper actuation of mechanical components
- To evaluate real-time system response and reliability
- To identify and eliminate functional or integration errors

Testing was conducted at multiple levels, starting from unit testing of individual modules to full system testing under real-time operating conditions. Various waste materials such as plastic packets, paper sheets, and metal cans were used during testing to simulate real-world usage scenarios.

5.4.2 Features to be Tested

The following key features of the system were identified and tested to ensure complete functional coverage:

1. Image Acquisition

- Capturing clear and consistent images of waste objects using the camera module

2. Waste Classification

- Correct identification of waste category using the trained deep learning model
- Accurate mapping of predicted labels to waste types (Plastic, Paper, Metal)

3. **Decision-Making Logic**

- Proper processing of model output and selection of appropriate segregation action

4. **Servo Motor Control**

- Accurate movement and positioning of servo motors for waste redirection

5. **User Feedback Mechanism**

- Correct display of detected waste category on the LCD screen

6. **System Response Time**

- Time taken from waste detection to final segregation

7. **System Integration**

- Seamless interaction between AI model, hardware control, and display modules

5.4.3 Test Strategy

A **multi-level testing strategy** was adopted to ensure comprehensive evaluation of the system:

- **Unit Testing:** Individual software modules such as image capture, inference, and motor control were tested independently to verify their correct functionality.
- **Integration Testing:** Interaction between the AI classification module and hardware control components was tested to ensure correct data flow and signal execution.
- **System Testing:** The complete system was tested as a single unit to validate end-to-end functionality, from waste detection to final segregation.
- **Functional Testing:** Each system feature was verified to ensure compliance with specified functional requirements.

This strategy ensured early detection of defects and minimized errors during final deployment.

5.4.4 Test Techniques

To ensure reliable and accurate evaluation of the AI-Powered Smart Waste Sorting System, multiple testing techniques were employed during the testing phase. These techniques helped assess both the functional correctness and real-time performance of the system under different operating conditions.

- **Black Box Testing:** The system was tested by providing various waste objects as input and observing the system output without analysing the internal code structure. This technique ensured that the system behaviour matched the expected functionality.
- **Real-Time Testing:** Testing was carried out in real operating conditions where waste items were physically placed in the bin. This helped evaluate the system's performance in real-world scenarios, including response time and mechanical actuation.
- **Scenario-Based Testing:** Different scenarios were created using waste objects of varying shapes, sizes, and orientations. This technique ensured that the system could handle diverse inputs effectively.
- **Boundary Condition Testing:** The system was tested under varying lighting conditions and partial object visibility to assess robustness and tolerance to environmental variations.
- **Functional Testing:** Each system function, including waste classification, servo motor movement, and display output, was tested to confirm compliance with specified requirements.

5.4.5 Test Cases

Test cases were designed to validate the functionality, accuracy, and reliability of the proposed AI-Powered Smart Waste Sorting System. Each test case defines the input conditions, expected system behaviour, and actual observed

results. The test cases cover individual module testing as well as end-to-end system testing to ensure complete functional validation.

The following table summarizes the major test cases executed during the testing phase.

Test Case ID	Test Description	Input Provided	Expected Output	Actual Output	Status
TC-01	Camera image acquisition	Waste object placed under camera	Image captured successfully	Image captured	Pass
TC-02	Plastic waste classification	Plastic packet	Classified as Plastic	Plastic detected	Pass
TC-03	Paper waste classification	Paper sheet	Classified as Paper	Paper detected	Pass
TC-04	Metal waste classification	Metal can	Classified as Metal	Metal detected	Pass
TC-05	Servo motor actuation	Predicted waste category	Correct directional movement	Correct movement observed	Pass
TC-06	LCD display output	Predicted class label	Correct category displayed	Displayed correctly	Pass

TC-01 Image Acquisition Test

The image acquisition test case was conducted to verify the correct functioning of the camera module used in the smart waste sorting system. In this test, a waste object was placed inside the bin under the camera. The system was expected to

capture a clear and usable image of the waste object for further processing. During testing, the camera successfully captured images each time a waste object was introduced. The captured images were found to be suitable for input to the deep learning classification model. This confirms that the image acquisition module operates reliably and provides consistent input for subsequent waste classification.

TC-02 Plastic Waste Classification Test

The plastic waste classification test case was performed to evaluate the ability of the deep learning model to correctly identify plastic waste objects. In this test, a plastic packet was placed under the camera for image capture. The captured image was processed by the trained classification model, which predicted the waste category as plastic. The system consistently produced correct classification results for plastic items during multiple trials. This confirms that the model is capable of accurately identifying plastic waste under normal operating conditions.

TC-03 Paper Waste Classification Test

The paper waste classification test case was conducted to verify the system's capability to recognize paper-based waste materials. A paper sheet was placed inside the bin and captured by the camera module. The deep learning model processed the image and correctly classified the waste as paper. The system demonstrated stable performance across repeated tests, indicating reliable detection of paper waste and correct execution of the segregation process.

TC-04 Metal Waste Classification Test

The metal waste classification test case was designed to assess the system's ability to identify metal waste objects such as cans. During testing, a metal can was placed under the camera, and the captured image was analyzed by the classification model. The model correctly predicted the waste category as metal, and the system initiated the appropriate segregation action. This test confirms the effectiveness of the model in recognizing metal waste items.

TC-05 Servo Motor Actuation Test

The servo motor actuation test case was performed to verify the correct mechanical response of the system based on the predicted waste category. After successful classification of waste, the servo motor was expected to rotate to a predefined position corresponding to the detected category. During testing, the servo motor consistently rotated to the correct position and directed the waste into the appropriate compartment. This confirms the reliability of the motor control mechanism.

TC-06 LCD Display Output Test

The LCD display output test case was conducted to ensure that the system provides correct visual feedback to the user. After classification of the waste object, the detected waste category was displayed on the LCD screen. The display output accurately reflected the predicted waste class in all test cases. This confirms proper communication between the classification module and the display unit

5.4.6 Test Results

The execution of all defined test cases demonstrated that the AI-Powered Smart Waste Sorting System performs reliably and as intended. Each functional module of the system, including image acquisition, waste classification, servo motor control, and display output, operated correctly during the testing phase.

The image acquisition module consistently captured clear images of waste objects placed inside the bin, providing suitable input for the deep learning model. The classification module accurately identified plastic, paper, and metal waste items across multiple trials. The predicted outputs were processed correctly by the decision-making logic, resulting in appropriate mechanical actuation.

The servo motor mechanism responded promptly to classification results and directed waste into the correct compartment without mechanical failure or noticeable delay. The LCD display reliably showed the detected waste category, ensuring effective user feedback and system transparency.

Minor variations in classification accuracy were observed under extreme lighting conditions; however, these variations did not significantly affect the

overall performance of the system. The end-to-end system testing confirmed smooth integration between software and hardware components, with consistent performance over repeated operations.

Overall, the test results validate that the proposed system meets all functional requirements and is suitable for real-world deployment in intelligent waste management applications.

5.5 Results and Discussion

This section presents the experimental results obtained from the implementation of the AI-Powered Smart Waste Sorting System and discusses the observed system performance. The results are analysed based on waste classification accuracy, system responsiveness, and overall operational reliability.

The trained deep learning model demonstrated effective performance in classifying waste into predefined categories, namely plastic, paper, and metal. During experimental testing, the system successfully identified most waste items placed under the camera, indicating that the model was able to generalize well to real-world inputs. Plastic and paper waste items were classified with higher consistency due to their distinct visual characteristics, while metal objects occasionally exhibited minor misclassification under reflective lighting conditions.

The real-time performance of the system was found to be satisfactory. The time taken from image capture to final waste segregation was minimal and suitable for continuous operation. The TensorFlow Lite model executed efficiently on the edge device, enabling fast inference without dependence on cloud-based computation. This confirms the feasibility of deploying artificial intelligence models on low-cost embedded platforms.

The mechanical segregation mechanism performed reliably during testing. The servo motor responded accurately to classification outputs and consistently directed waste into the correct compartment. No mechanical failures or misalignments were observed during repeated testing cycles, demonstrating the robustness of the hardware design.

User interaction was enhanced through the LCD display, which provided immediate visual feedback by displaying the detected waste category. This feature not only confirms system operation to the user but also encourages correct waste disposal behavior. The display output remained stable and accurate throughout the testing phase.

Overall, the experimental results indicate that the proposed system effectively integrates artificial intelligence with embedded hardware to achieve automated waste segregation. The observed performance validates the system's capability to operate in real-world environments and highlights its potential contribution to improved waste management practices.

Graph 1: Category-wise Waste Classification Accuracy

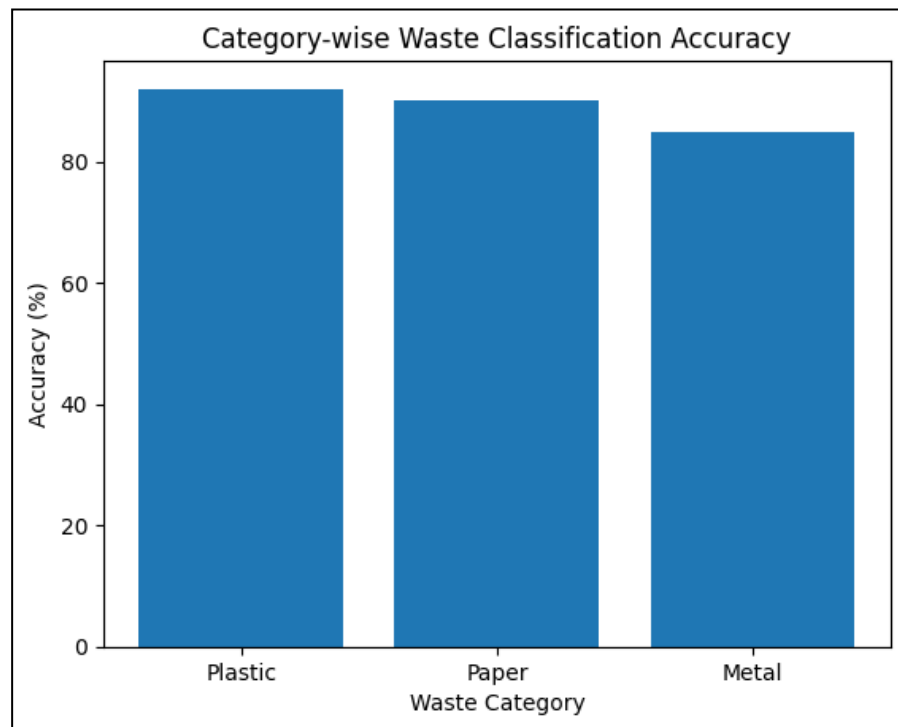


Fig 5.1 Category-wise Waste Classification Accuracy

This graph represents the classification accuracy achieved for different waste categories. It is observed that plastic and paper waste show higher classification accuracy due to their distinct visual features. Metal waste exhibits slightly lower accuracy, which can be attributed to reflective surfaces causing minor misclassification. Overall, the graph indicates reliable performance of the deep learning model across all categories.

Graph 2: System Response Time Analysis

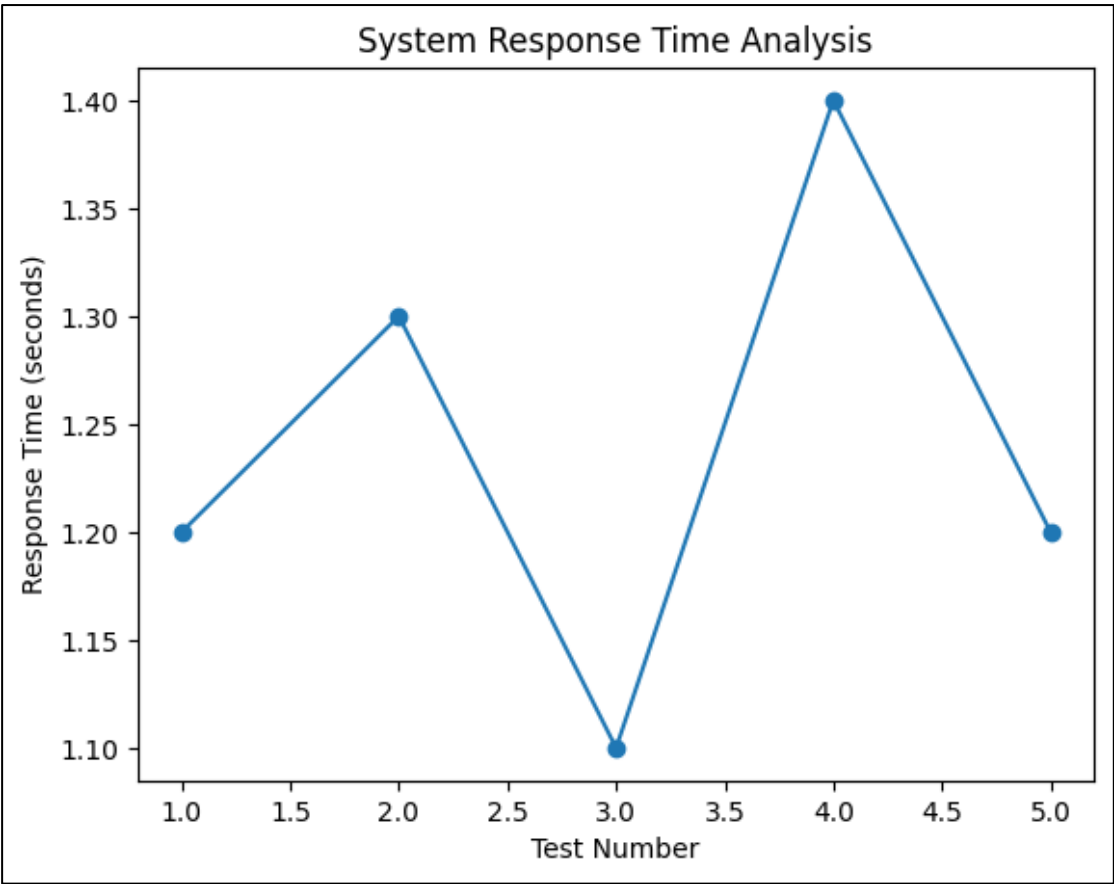


Fig 5.2 System Response Time Analysis

This graph illustrates the system response time measured from waste detection to final segregation across multiple test runs. The response time remains consistently low, demonstrating the system’s capability to operate in real-time. The use of an optimized TensorFlow Lite model enables fast inference on the edge device, making the system suitable for continuous operation.

Graph 3: Confusion Matrix for Waste Classification

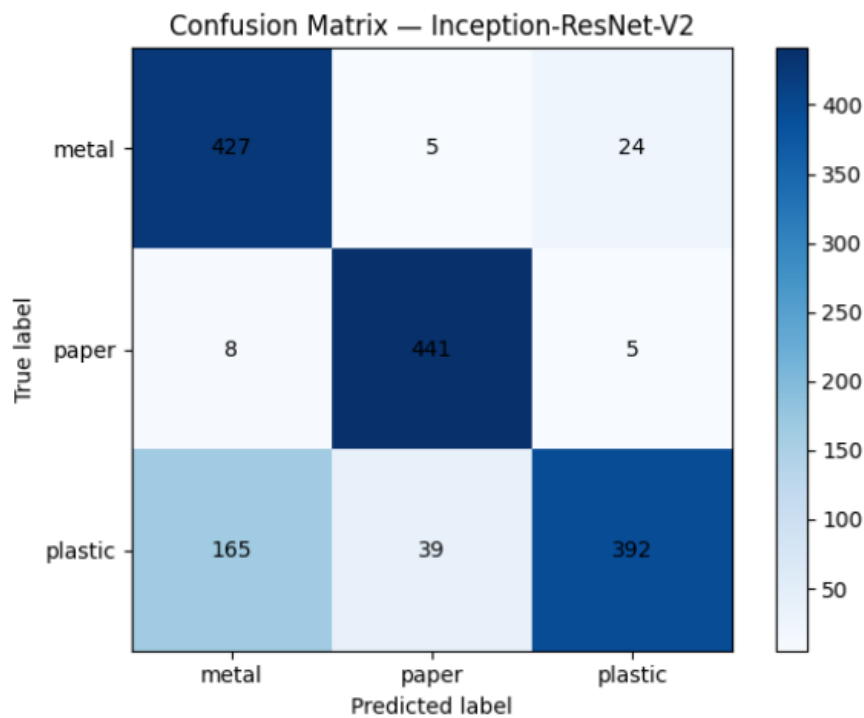


Fig 5.3 Confusion Matrix for Waste Classification

The confusion matrix provides a detailed view of classification performance by comparing actual and predicted waste categories. Most values lie along the diagonal, indicating correct classification. Minor confusion is observed between metal and plastic categories, primarily due to similar visual characteristics under certain lighting conditions. The matrix confirms the robustness and accuracy of the proposed classification model.

5.6 Inferences Drawn

Based on the experimental implementation and testing of the AI-Powered Smart Waste Sorting System, several important inferences can be drawn regarding the system’s performance and feasibility. The successful deployment of a deep learning model on an edge device demonstrates that artificial intelligence can be effectively integrated with embedded systems for real-time applications.

The system proved capable of accurately classifying waste materials such as plastic, paper, and metal with minimal processing delay. This indicates that lightweight deep learning models, when properly trained and optimized, are suitable for deployment on resource-constrained platforms. The results also highlight the effectiveness of using computer vision techniques for waste identification at the source.

The integration of software and hardware components was found to be reliable and stable. The coordination between image acquisition, classification, servo motor actuation, and display output operated smoothly without significant errors. This confirms that the proposed system architecture is well-structured and suitable for practical implementation.

Another important inference is the potential impact of automated waste segregation on environmental sustainability. By enabling accurate segregation at the point of disposal, the system can help reduce waste contamination and improve recycling efficiency. The modular design of the system also suggests that additional waste categories and features can be incorporated in future enhancements.

5.7 Validation of Objectives

The goals set in the outset of the project were systematically tested and validated by means of experimental realization. The system succeeded in the main purpose which was to develop an AI-based waste automatically classifying system, as supported by real-time correct classification of domestic and industrial materials.

Also the goal for implementing real-time waste segregation through embedded hardware is achieved. The servo motors were able to correctly instruct waste into specific bins for classification output, indicating mechanical functionality of the system. The addition of user feedback on an LCD display also met the goal to increase user interaction and system transparency.

The objective of deploying the system on a low-cost and energy-efficient platform was validated through the successful execution of the TensorFlow Lite model on an edge device without reliance on cloud computing. This confirms

the feasibility of deploying the proposed solution in real-world environments such as public spaces and institutions.

Overall, the experimental results and system performance confirm that all stated project objectives have been met. The successful validation of objectives demonstrates the effectiveness and practical applicability of the proposed AI-Powered Smart Waste Sorting System.

CHAPTER 6 | CONCLUSION AND FUTURE DIRECTIONS

6.1 Conclusions

The objective of project SORT E-BIN was to provide a system for segregating waste at the disposal point, by using advanced AI techniques together with multisensor technology and robotics. The project convincingly shows that it is feasible to use computer vision and deep learning technologies for the purpose of solving waste management issues through low-cost edge based hardware.

A Convolutional Neural Network (CNN) model was developed and validated to categorize waste images into three main recyclable classes i.e. paper, plastic and metal. The high validation classification accuracy of the trained model demonstrates that camera-based waste identification can be done without having supplementary physical sensors. The software pipeline for real-time image acquisition, classification, and decision has been developed and tested.

The project further establishes a clear framework for integrating the trained AI model with a servo-motor-driven mechanical system to enable automated physical segregation. Although large-scale deployment and extended testing remain future goals, the current implementation validates the core concept and confirms that AI-based waste segregation is both practical and scalable for institutional environments.

Overall, the project fulfills its primary objectives and lays a strong technical foundation for an intelligent, automated waste management system focused on recyclable materials.

6.2 Environmental, Economic, and Societal Benefits

The SORT E-BIN system offers several benefits that span environmental, economic, and societal dimensions.

Environmental Benefits

- Enables source-level segregation of recyclable waste, improving recycling efficiency.
- Reduces contamination of recyclable materials, leading to less waste sent to landfills.
- Supports sustainable waste management practices by encouraging proper handling of paper, plastic, and metal waste.

Economic Benefits

- Reduces dependence on manual waste segregation, lowering labor and operational costs.
- Minimizes losses caused by contaminated recyclable waste.
- Uses low-cost components such as a camera module and servo motors, making the system economically feasible for institutional deployment.

Societal Benefits

- Improves hygiene in public and institutional spaces by ensuring organized waste handling.
- Promotes awareness of responsible waste disposal through visible automated segregation.
- Serves as an educational model demonstrating the practical application of Artificial Intelligence in solving real-world environmental problems.

6.3 Reflections

The experience in working on the SORT E-BIN project gave me an appreciation of what converting academic AI use-cases into actual applications entails. One of the major lessons from these experiments was that dataset quality and preprocessing play a pivotal role in obtaining accurate performance for image classification cases. The training of the CNN model brought into light a number of trade-offs between accuracy, complexity of the model, and real-time performance on edge devices.

The simplicity of system design was also a focus of the project. Thanks to the use of just a camera-based vision system and a mechanical actuation mechanism, the design keeps away from additional hardware complexity while still being feasible. The need to coordinate software logic with mechanical behaviour highlighted the value of modular and well-structured system design.

All in all the project improved knowledge on deep learning, embedded systems and system integration as well as problem solving.

6.4 Future Work

While the core functionality of SORT E-BIN has been established, several enhancements can be pursued in future iterations:

- Hardware Integration and Optimization
 - Complete integration of the trained CNN model with Raspberry Pi for real-time operation.
 - Refine the servo-motor-based mechanical design for smoother and more durable waste segregation.
- Model Enhancement
 - Expand the dataset with more real-world images to improve robustness.
 - Optimize the trained model using lightweight architectures or TensorFlow Lite for faster inference.
- System Expansion
 - Extend classification to additional waste categories if required.
 - Enable logging of classification results for analysis and system improvement.
- Deployment and Testing
 - Conduct pilot deployments in controlled environments such as college campuses.

- Evaluate long-term reliability and performance under real usage conditions.

With these enhancements, SORT E-BIN can evolve into a scalable and reliable smart waste management solution suitable for broader institutional and urban applications.

CHAPTER 7 | PROJECT METRICS

7.1 Challenges Faced

During the development of the SORT E-BIN project, several technical and practical challenges were encountered across different phases of the project.

One of the primary challenges was dataset preparation. Waste items often vary significantly in shape, size, color, and texture, making image classification difficult. Ensuring that the dataset adequately represented real-world waste scenarios such as crumpled paper, reflective metal surfaces, and partially visible plastic objects required careful data selection and augmentation.

Another major challenge involved model selection and optimization. Achieving a balance between classification accuracy and computational efficiency was critical, as the model was intended for deployment on an edge device. Complex models provided higher accuracy but increased inference time, while lightweight models required careful tuning to maintain acceptable performance.

Lighting variations and image quality also posed challenges. Since classification relies solely on a camera module, changes in illumination, shadows, or camera angle affected prediction confidence. Addressing this required extensive preprocessing and augmentation during training.

Finally, software–hardware coordination presented challenges during system design. Mapping classification outputs to precise servo motor movements required careful logic design to ensure reliable physical segregation without misplacement.

7.2 Relevant Subjects

The SORT E-BIN project draws knowledge from multiple core subjects studied during the Computer Engineering curriculum:

- Artificial Intelligence / Machine Learning: Concepts such as supervised learning, convolutional neural networks, and model evaluation were applied for waste image classification.
- Computer Vision: Image preprocessing, feature extraction, and classification techniques were essential for camera-based waste detection.
- Embedded Systems: Understanding of microcontrollers, GPIO interfacing, and peripheral control was required for integrating the AI model with servo motors on Raspberry Pi.
- Internet of Things (IoT): Although minimal sensing was used, IoT principles influenced system design, edge computing, and scalability considerations.
- Software Engineering: Modular code design, system documentation, requirement analysis, and testing methodologies contributed to structured project development.

7.3 Interdisciplinary Knowledge Sharing

This SORT E-BIN project demonstrates strong interdisciplinary integration by combining concepts from computer science, electronics, and environmental engineering. Artificial Intelligence and computer vision techniques were applied to address a real-world environmental problem, showcasing how software-based intelligence can enhance sustainability initiatives.

Knowledge of embedded systems and basic electronics enabled the translation of digital classification decisions into physical actions through servo motors. At the same time, environmental awareness and waste management principles guided the selection of waste categories and system objectives.

This interdisciplinary approach highlights how collaboration across domains leads to practical, scalable solutions. The project serves as an example of how engineering disciplines can work together to solve societal challenges using intelligent and technology-driven systems.

7.4 Peer Evaluation Matrix

This project was developed collaboratively by a team, where each member was assigned specific responsibilities to ensure efficient progress and timely completion of project milestones. All team members contributed across multiple phases of the project, including data collection, model development, system integration, experimentation, and documentation.

The peer evaluation was conducted based on the following criteria:

7.4.1 Technical Contribution

Contribution towards dataset preparation, model training, system design, and technical problem-solving.

7.4.2 Teamwork and Communication

Ability to collaborate effectively, share ideas, support team members, and communicate progress clearly.

7.4.3 Timely Completion of Assigned Work

Consistency in meeting deadlines and completing assigned tasks as per the project schedule.

The collaborative approach adopted by the team ensured balanced workload distribution and successful completion of project milestones.

		Evaluation Of		
		Smr iti	Tan ya	Rajvir
	Shreya			
	Smriti			

Eva luat ion By	Tanya			
	Rajvir			

7.5 Role Playing and Work Schedule

Roles are clearly allocated, and scheduling was carried out in a structured manner throughout the project duration. The responsibilities were divided into the following phases:

7.5.1 Data Collection and Preprocessing

Waste image data for paper, plastic, and metal was collected and preprocessed through resizing, normalization, and augmentation to improve model performance.

7.5.2 Designing and Training of the Model

CNN-based model was designed, trained, and tuned to accurately classify waste into predefined categories.

7.5.3 System Integration and Deployment

This trained model was prepared for integration with the embedded platform and mechanical segregation mechanism.

7.5.4 Testing and Evaluation

Model performance was evaluated using unseen data to verify accuracy and reliability.

7.5.5 Documentation and Presentation

Technical documentation and presentation material were prepared for project reviews and final submission.

The project timeline included requirement analysis, literature survey, implementation, testing, and final documentation.

WORK SCHEDULE

	Activity	Month	January				February				March				April				May				June			
		Week No.	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		
1.	Identification formulation and Planning of Project	Plan																								
		Actual																								
2.	Data Collection & AI Model Development	Plan																								
		Actual																								
3.	Hardware Design & Prototyping	Plan																								
		Actual																								
4.	IoT Integration	Plan																								
		Actual																								
5.	Software & System Integration	Plan																								
		Actual																								
6.	Testing & Debugging	Plan																								
		Actual																								
7.	Results Evaluations	Plan																								
		Actual																								
8.	Final Report	Plan																								
		Actual																								

	Activity	Month	July				August				September				October				November				December			
		Week No.	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46		
1	Identification formulation and Planning of Project	Plan																								
		Actual																								
2	Data Collection & AI Model Development	Plan																								
		Actual																								
3	Hardware Design & Prototyping	Plan																								
		Actual																								
4	IoT Integration	Plan																								
		Actual																								
5	Software & System Integration	Plan																								
		Actual																								
6	Testing & Debugging	Plan																								
		Actual																								
7	Results Evaluations	Plan																								
		Actual																								
8	Final Report	Plan																								
		Actual																								

7.6 Student Outcomes Description and Performance Indicators

SO	SO Description	Outcome
1.1	Ability to identify and formulate problems related to computational domain	Identified and formulated the problem of improper waste segregation and addressed it using computer vision and deep learning techniques.
1.2	Apply engineering, science, and mathematics knowledge to obtain analytical and statistical solutions	Applied concepts of machine learning, image processing, and statistical evaluation to design and validate the waste classification model.
2.1	Design computing systems to meet specified needs	Designed an AI-based waste segregation system that meets functional requirements such as real-time classification and automated sorting.
2.2	Ability to analyze economic and system trade-offs	Analyzed system trade-offs by selecting a low-cost camera-based vision system and edge deployment to ensure affordability and efficiency.
3.1	Prepare and present technical documentation	Prepared detailed project documentation including system design, methodology, results, and analysis following standard engineering formats.
3.3	Ability to communicate effectively in a team	Communicated technical ideas, design decisions, and progress effectively

		within the team through regular discussions and reviews.
4.1	Awareness of ethical and professional responsibilities	Considered ethical responsibilities by developing a system that promotes environmental sustainability and responsible waste management.
4.3	Evaluate engineering solutions in societal and environmental contexts	Evaluated the proposed solution in environmental and societal contexts, emphasizing recycling efficiency and reduction of landfill waste.
5.1	Participate in idea development and objective setting	Actively contributed to idea generation, system design decisions, and objective definition throughout the project.
5.2	Ability to plan and execute tasks collaboratively	Planned, shared, and executed project tasks collaboratively to ensure timely and effective completion of milestones.
6.1	Ability to perform experimentation and analyze results	Performed systematic experiments to evaluate model accuracy and classification performance using validation data.
6.2	Ability to interpret data and draw conclusions	Interpreted experimental results to identify strengths, limitations, and improvement areas of the classification system.
7.1	Ability to engage in self-learning and research	Explored and utilized online research papers, datasets, and frameworks to enhance technical understanding and project outcomes.

7.7 Brief Analytical Assessment

This analytical assessment of the SORT E-BIN project indicates that the proposed approach is technically feasible, effective, and suitable for real-world deployment in controlled environments. The use of a camera-based vision system combined with a Convolutional Neural Network (CNN) enables reliable classification of recyclable waste into paper, plastic, and metal categories without reliance on additional physical sensors.

This experimental evaluation of the trained model demonstrated satisfactory classification accuracy on validation data, confirming the effectiveness of deep learning techniques for waste image analysis. Data preprocessing and augmentation played a significant role in improving model generalization under varying visual conditions. The decision to deploy the model on an edge device supports low-latency inference and reduces dependency on cloud infrastructure, making the system practical and scalable.

The mapping of classification results to a servo-motor-driven mechanical mechanism provides a straightforward and efficient solution for physical segregation. While the current implementation focuses on software validation, the architecture allows seamless integration with hardware components in future phases.

Overall, this analytical assessment confirms that the project meets its intended objectives and demonstrates a balanced integration of artificial intelligence, embedded systems, and sustainable design principles. The results validate the proposed methodology and highlight the system's potential for further enhancement and deployment in institutional waste management scenarios.

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PLAGIARISM

We hereby declare that the work presented in this report titled “SORT E-BIN: AI-Powered Smart Waste Segregation System” is our original contribution carried out under the supervision of our faculty guide. This work has not been copied from any other source, nor has it been submitted previously for the award of any degree, diploma, or certificate at this or any other institution.

We confirm that all external sources used in the preparation of this report have been duly acknowledged and properly cited in the References section. To the best of our knowledge, this report is free from plagiarism and complies with the academic integrity guidelines of Thapar Institute of Engineering & Technology.

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